

MinerU2.5-Pro: Pushing the Limits of Data-Centric Document Parsing at Scale

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Current document parsing methods advance primarily through model architecture innovation, while systematic engineering of training data remains underexplored. Yet state-of-the-art models spanning diverse architectures and parameter scales exhibit highly consistent failure patterns on the same set of hard samples, suggesting that the performance bottleneck stems from shared deficiencies in training data rather than from architectural differences. Building on this finding, we present MinerU2.5-Pro, which advances the state of the art purely through data engineering and training strategy design while retaining the 1.2B-parameter architecture of MinerU2.5 unchanged. At its core is a Data Engine co-designed around coverage, informativeness, and annotation accuracy: Diversity-and-Difficulty-Aware Sampling expands training data from under 10M to 65.5M samples while mitigating distribution shift; Cross-Model Consistency Verification leverages output consensus among heterogeneous models to assess sample difficulty and generate reliable annotations; the Judge-and-Refine pipeline improves annotation quality for hard samples through render-then-verify iterative correction. A three-stage progressive training strategy—large-scale pre-training, hard sample fine-tuning, and GRPO alignment—sequentially exploits these data at different quality tiers. On the evaluation front, we rectify element-matching biases in OmniDocBench v1.5 and introduce a Hard subset, establishing the more discriminative OmniDocBench v1.6 protocol. Without any architectural modification, MinerU2.5-Pro achieves 95.69 on OmniDocBench v1.6, improving over the same-architecture baseline by 2.71 points and surpassing all existing methods, including those based on models with over 200× more parameters.

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Code: <https://github.com/opensdata/MinerU>

Model: <https://huggingface.co/opensdata/MinerU2.5-Pro-2604-1.2B>

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Contents

1	Introduction	3
2	Related Work	4
2.1	Document Parsing Methods	4
2.2	Data-Centric AI	5
2.3	Document Parsing Evaluation	5
3	Data Engine	6
3.1	Diversity-and-Difficulty-Aware Data Sampling	7
3.2	Cross-Model Consistency Verification	8
3.3	Annotation Pipeline for Hard Case	9
4	Progressive Training Strategy	10
4.1	Stage 1: Document Parsing Pre-training	11
4.2	Stage 2: High-Quality Supervised Fine-Tuning	11
4.3	Stage 3: Reinforcement Learning with GRPO	11
5	OmniDocBench v1.6	12
5.1	Motivation	12
5.2	Multi-Granularity Adaptive Matching	13
5.3	Hard Subset and Three-Tier Evaluation Protocol	14
6	Experiments	14
6.1	Evaluation Setup	14
6.2	End-to-End Document Parsing	15
6.3	Element-Specific Parsing	16
7	Conclusion	17
A	Prompt Design and Task Examples	23
A.1	Layout Detection	23
A.2	Text Recognition	25
A.3	Formula Recognition	25
A.4	Table Recognition	25
A.5	Image-Aware Parsing	28
B	Extended Parsing Capabilities	28
C	OmniDocBench v1.6 Detailed Results	34
C.1	Detailed Results on Base Subset	34
C.2	Detailed Results on Hard Subset	34
D	Qualitative Comparison with SOTA Methods	34
D.1	Table Recognition	36
D.2	Formula Recognition	36
D.3	Image-Aware Parsing	36

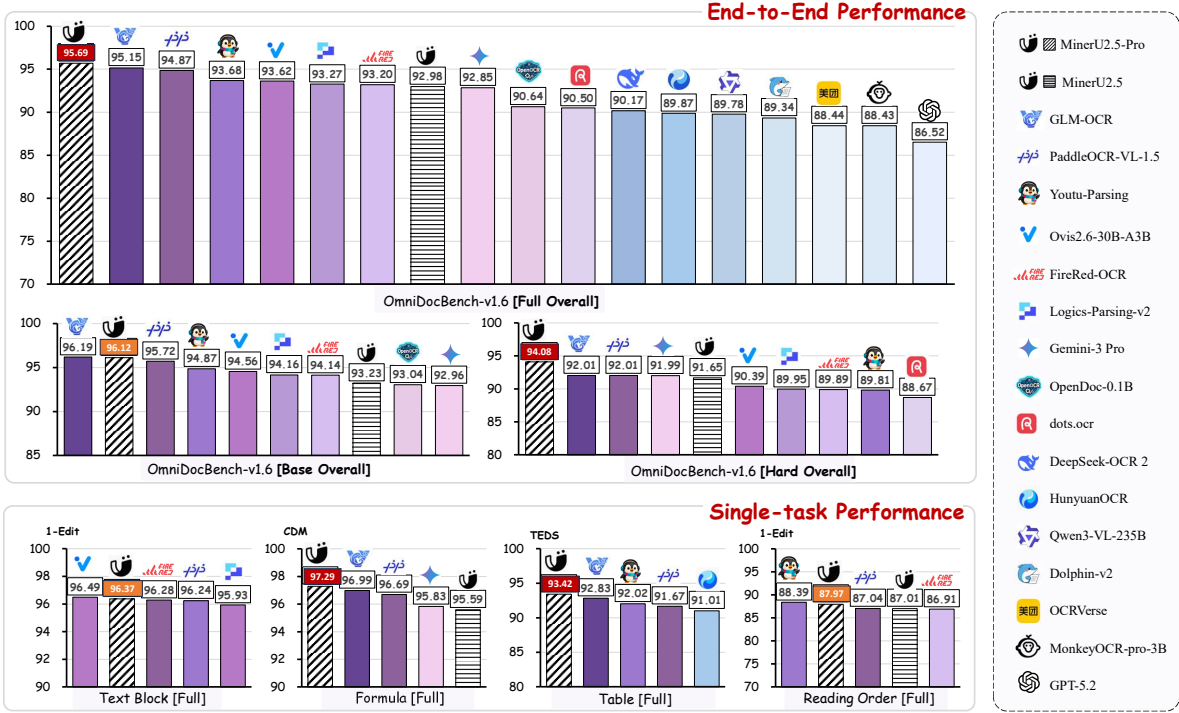


Figure 1: Performance comparison on OmniDocBench v1.6, which comprises Base (standard samples), Hard (challenging samples), and Full (overall) splits. Built upon MinerU2.5 [25] with its 1.2B-parameter architecture entirely unchanged, MinerU2.5-Pro improves the overall score from 92.98 to 95.69 purely through data engineering and training strategy design, outperforming both specialized document parsing models (e.g. GLM-OCR [11], PaddleOCR-VL-1.5 [8], Youtu-Parsing[45]) and general-purpose VLMs (e.g. Gemini 3 Pro, Qwen3-VL-235B [43]). Detailed results are presented in Table 2.

1 Introduction

Document parsing converts unstructured documents such as PDFs into structured, machine-readable formats (e.g. Markdown), serving as critical infrastructure for LLM training data pipelines [49, 29, 35] and retrieval-augmented generation systems [15, 37, 48]. As end-to-end approaches based on vision-language models (VLMs) progressively replace traditional pipeline systems [2, 39, 25], research has focused predominantly on architectural innovation and inference efficiency, leading to rapid score convergence among top models on standard benchmarks.

Yet this convergence raises a deeper question: what constitutes the remaining performance bottleneck? Our cross-analysis of parsing results from multiple state-of-the-art models—spanning diverse architectures and parameter scales—on large-scale real-world PDFs reveals a striking pattern: these models exhibit highly similar failure modes on the same hard samples, with certain parsing errors common to all tested systems. Since these systematic failures transcend any particular architecture, they point to a common root cause: the current performance bottleneck in document parsing stems primarily from shared deficiencies in training data, not from model architecture itself.

This data bottleneck manifests in two interrelated dimensions. First, *insufficient coverage*: for instance,

MinerU2.5’s training data totals less than 10M pages with distributions concentrated on high-frequency categories, severely underrepresenting long-tail scenarios such as complex nested tables and dense formula layouts. Second, an *annotation quality paradox*: the hard samples that contribute most to model improvement are precisely those for which automatic annotation is least reliable, since no mainstream model can consistently parse them correctly. Structural annotations for complex tables and L^AT_EX transcriptions of dense formulas are highly error-prone, and this annotation noise propagates directly into model behavior during supervised fine-tuning. These two issues are deeply intertwined: simply scaling data volume is insufficient to raise the performance ceiling, as added data only amplifies existing distribution biases and annotation noise.

Beyond data, our cross-analysis also exposes blind spots in the existing evaluation framework. OmniDocBench v1.5 contains relatively few hard samples, and its element-matching logic exhibits systematic biases toward specific output formats, introducing scoring artifacts that complicate fair cross-system comparison. We accordingly introduce OmniDocBench v1.6, which corrects these matching biases and incorporates a dedicated Hard subset to establish a Base/Hard/Full three-tier evaluation protocol.

Based on this analysis, we argue that as model architectures mature, systematic data engineering becomes the primary lever for advancing document parsing performance. To test this hypothesis, we build MinerU2.5-Pro—retaining the identical 1.2B-parameter decoupled coarse-to-fine architecture of MinerU2.5 [25] and focusing all optimization on the Data Engine and training strategy, ensuring that all performance gains are attributable to data-level improvements. On OmniDocBench v1.6, MinerU2.5-Pro achieves 95.69 (baseline 92.98, +2.71), surpassing all existing methods, including models with over 200× more parameters (Figure 1).

Our contributions are summarized as follows:

- A Data Engine co-designed around coverage, informativeness, and annotation accuracy. It comprises three core components—Diversity-and-Difficulty-Aware Sampling (DDAS), Cross-Model Consistency Verification (CMCV), and a Judge-and-Refine annotation pipeline—that together expand training data from under 10M to 65.5M pages while systematically improving annotation quality through a closed-loop progression from sampling to refinement.
- A three-stage progressive training strategy—large-scale pre-training, high-quality hard sample fine-tuning, and GRPO format alignment—matched to the data quality tiers produced by the Data Engine. With these data and training improvements alone, the same 1.2B-parameter model achieves state-of-the-art performance on OmniDocBench v1.6, surpassing all existing methods.
- OmniDocBench v1.6, an upgraded evaluation protocol that corrects element-matching biases in v1.5 through Multi-Granularity Adaptive Matching and introduces a Hard subset, establishing a Base/Hard/Full three-tier framework for fairer and more discriminative evaluation.

2 Related Work

2.1 Document Parsing Methods

Existing document parsing methods fall into three paradigms based on system architecture.

Pipeline-based methods. These methods decompose document parsing into independent subtasks—layout detection, text recognition, table extraction, formula recognition—and execute them in a cascade [35, 50, 21, 28, 7]. This modular design enables independent optimization of each component but suffers from error propagation and inter-module information loss.

End-to-end VLM methods. These methods directly map document images to structured output, avoiding the cascading errors inherent in pipeline approaches. Nougat [2], built on the Donut architecture [16], established a strong baseline for the image-to-markup paradigm on academic documents; GOT-OCR 2.0 [39] unified scene text and document OCR within a single model. Subsequent works such as Ocean-OCR [3], olmOCR [29], and dots.ocr [18] employ native-resolution visual encoders to further improve performance. However, native-resolution processing incurs $\mathcal{O}(N^2)$ token complexity, creating efficiency bottlenecks for high-resolution documents.

Decoupled VLM methods. These methods separate layout analysis from content recognition, combining the controllability of pipeline approaches with the semantic modeling power of VLMs. Early works such as Dolphin [12] and MonkeyOCR [19] demonstrated the viability of this paradigm but faced limitations in resolution handling or system complexity. MinerU2.5 [25] unifies layout analysis and content recognition within a single 1.2B-parameter model with native-resolution support [26], balancing resolution fidelity, efficiency, and deployment complexity. Subsequent works extend the decoupled paradigm along various axes: multi-token prediction for throughput [11], diffusion-based decoding for improved parsing efficiency and robustness [9], non-planar document handling [8], in-the-wild robustness [32], and high-compression vision-text mappings [40]. General-purpose VLMs such as Gemini 2.5 Pro [5] and Qwen2.5-VL-72B [1] also achieve competitive results, though their large parameter scales hinder cost-effective deployment at production scale.

Across these works, the main line of methodological evolution focuses on architecture design and inference efficiency, while systematic engineering of training data—co-optimizing coverage, informativeness, and annotation accuracy—has not been adequately explored as an independent research problem. Our work addresses this dimension and is largely complementary to the architectural advances above.

2.2 Data-Centric AI

The Data-Centric AI paradigm [24, 47] advocates systematically improving data quality while keeping the model fixed, and has been validated in vision-language pretraining [14] and large language model fine-tuning [53]. In document parsing, however, data engineering remains fragmented: olmOCR [29] emphasizes data scale expansion over quality stratification; DocGenome [42] is restricted to academic papers and lacks difficulty differentiation; existing technical reports [25, 11, 8] describe training data but treat it as a prerequisite for model training rather than an independent research subject.

Our work treats data construction for document parsing as a standalone systematic research problem, co-optimizing coverage, informativeness, and annotation accuracy within a unified framework. Methodologically, our CMCV approach draws on the core principles of ensemble-based active learning [30] and query-by-committee [13] by leveraging multi-model disagreement to quantify sample informativeness. Beyond standard disagreement-based selection, CMCV couples difficulty information with downstream annotation strategies in a closed loop and addresses the document-parsing-specific challenge of unreliable hard sample annotation through the Judge-and-Refine pipeline.

2.3 Document Parsing Evaluation

Document parsing evaluation involves both metric design and evaluation protocol. At the metric level, text recognition commonly uses edit distance [17], table structure recovery uses TEDS [51], and formula recognition has recently shifted from BLEU to CDM (Character Detection Matching) [36]. OmniDocBench [27] integrates all three metrics and provides one of the most comprehensive document parsing evaluation frameworks to date; OCRBench [20] and CC-OCR [44] focus on evaluating multimodal models’ overall OCR capabilities.

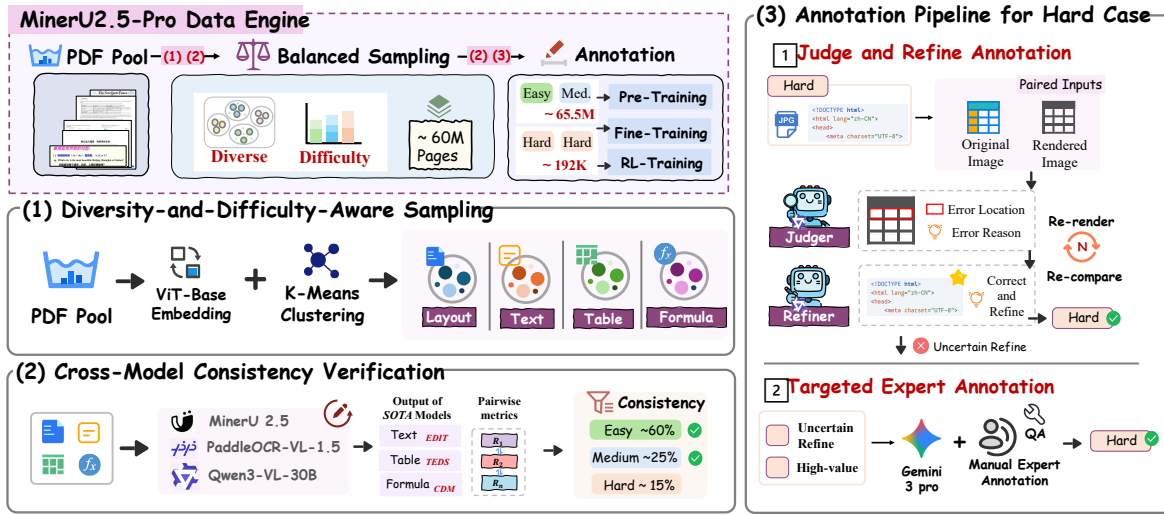


Figure 2: Overview of the Data Engine pipeline. The system co-optimizes three dimensions—Coverage, Informativeness, and Accuracy—through three synergistic stages: Diversity-and-Difficulty-Aware Sampling (DDAS), Cross-Model Consistency Verification (CMCV), and the Annotation Pipeline for Hard Case.

At the protocol level, however, the critical impact of element-matching strategies on evaluation fairness remains largely overlooked. End-to-end systems vary in output granularity, segmentation strategies, and format conventions, and the choice of matching algorithm systematically affects evaluation scores. We identify such systematic biases in OmniDocBench v1.5 and rectify them in v1.6 through Multi-Granularity Adaptive Matching (detailed in [Section 5](#)).

3 Data Engine

To address the data deficiencies identified above, we first examine the limitations of existing data pipelines. MinerU2.5 [25] built a data pipeline comprising cluster-based sampling, Iterative Model Inference Consistency (IMIC) hard sample mining, and model-based annotation refinement, but these components operate independently without joint optimization of coverage, informativeness, and accuracy: sampling is not informed by difficulty, annotation refinement applies a uniform strategy regardless of sample difficulty, and hard samples mined by IMIC still face unreliable automatic annotation. Similar limitations exist in PaddleOCR-VL-1.5’s Uncertainty-Aware Cluster Sampling (UACS) [8].

The Data Engine of MinerU2.5-Pro is co-designed around these three dimensions. DDAS expands data coverage through task-aware clustering and mitigates distribution shift ([Section 3.1](#)). CMCV performs difficulty stratification on the sampled data via multi-model cross-validation, identifying highly informative samples ([Section 3.2](#)). The Annotation Pipeline for Hard Case improves annotation accuracy through render-then-verify iterative correction, with residual samples beyond automatic correction routed to targeted expert annotation to guarantee final quality ([Section 3.3](#)). Together, these components form a coarse-to-fine quality progression, enabling simultaneous data scaling (under 10M $\rightarrow$ 65.5M) and annotation quality improvement. The overall pipeline is illustrated in [Figure 2](#).

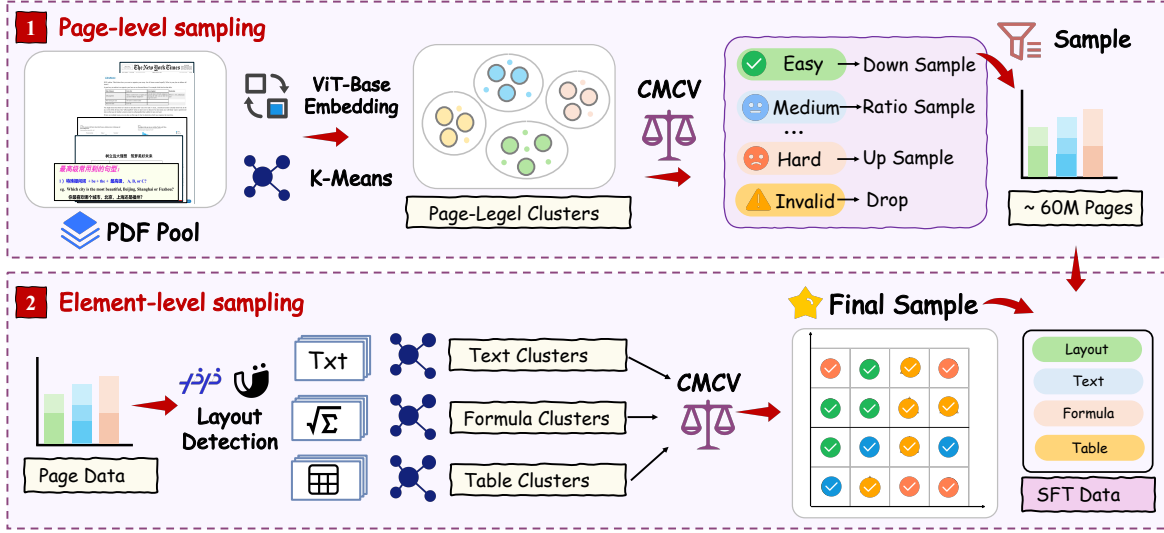


Figure 3: The DDAS pipeline operates at two granularity levels. **Upper:** Page-level sampling for layout detection data—pages from the PDF pool are embedded via ViT-base, clustered, and resampled by jointly weighting cluster diversity and CMCV-derived difficulty, yielding about 60M pages with balanced distribution and difficulty coverage. **Lower:** Element-level sampling—the selected pages are parsed by layout detection models into text, formula, and table blocks; each element type is independently clustered and assessed by CMCV, then sampled to balance both diversity and difficulty at the element granularity. The two levels are combined to produce the final training data for layout, text, formula, and table subtasks.

3.1 Diversity-and-Difficulty-Aware Data Sampling

Training data for document parsing exhibits a typical long-tail distribution problem: high-frequency categories (*e.g.* standard academic papers, single-column reports) dominate the data pool, while long-tail scenarios such as complex nested tables, dense formula layouts, and unconventional multi-column layouts are severely underrepresented. As noted above, existing approaches [25, 8] rely on single-model signals for difficulty estimation, which cannot distinguish model-specific weaknesses from universally hard samples.

We propose Diversity-and-Difficulty-Aware Sampling (DDAS), which jointly optimizes diversity and difficulty at both page and element granularity. Central to DDAS is Cross-Model Consistency Verification (CMCV, detailed in Section 3.2), which leverages prediction agreement among heterogeneous models to classify samples into Easy/Medium/Hard difficulty tiers. The overall pipeline is shown in Figure 3.

Stage 1: Page-level sampling. Pages in the document pool are embedded using ViT-base features (512-dim) and grouped via K-Means clustering. An initial uniform sample from each cluster is then evaluated by page-level CMCV (Section 3.2) to obtain difficulty labels. Based on the resulting difficulty distribution within each cluster, sampling weights are adjusted: clusters dominated by Easy samples receive lower weight, clusters with diverse difficulty distributions receive higher weight, and clusters dominated by invalid content (non-target languages, blank pages, *etc.*) are filtered out. Using the adjusted weights, we expand sampling from the original document pool to obtain the full page-level candidate set with CMCV difficulty annotations.

Stage 2: Element-level sampling. From the page-level candidate set, we extract individual elements (text, formula, and table blocks) using MinerU2.5 and PaddleOCR-VL layout detection models. For each element type, visual features are extracted and clustered independently, while element-level CMCV assigns difficulty labels. At this stage, all four subtasks—layout, text, formula, and table—have annotations along both the diversity (clustering) and difficulty (CMCV) dimensions.

Final sampling. Balanced sampling is performed in the joint cluster-difficulty space across all four subtasks: along the diversity dimension, large clusters are downsampled and small clusters are upsampled to correct long-tail shift; along the difficulty dimension, Medium and Hard samples are upweighted to enhance training signal informativeness. The final output is an SFT training set that covers all subtasks and balances diversity with difficulty.

By coupling clustering with CMCV at both page and element granularity, DDAS enables sampling decisions to simultaneously account for data distribution and training value, maximizing training signal density while controlling total data volume.

3.2 Cross-Model Consistency Verification

DDAS relies on difficulty labels to guide sampling weight allocation, and subsequent annotation refinement and expert annotation also require difficulty information to determine resource investment strategies. However, ground truth is unavailable for massive unlabeled data. IMIC in MinerU2.5 [25] and UACS in PaddleOCR-VL-1.5 [8] use output consistency from multiple inferences of a single model as a difficulty proxy. This paradigm captures only the epistemic uncertainty of a single model and cannot distinguish between model-specific blind spots and universally hard problems—the former can be directly rectified via cross-model consensus, while the latter necessitates additional quality refinement or even human intervention. This distinction is critical for annotation strategy selection.

We propose Cross-Model Consistency Verification (CMCV), which extends difficulty assessment from single-model introspection to multi-model cross-validation. The underlying premise is intuitive and empirically supported: when multiple heterogeneous models produce consistent outputs for a given sample, the result is highly likely to be correct; when all models diverge substantially, the sample is genuinely difficult and none of the models can parse it reliably. Based on this premise, we run three heterogeneous document parsing models (MinerU2.5 [25], PaddleOCR-VL [6], Qwen3-VL-30B [43]) independently on the candidate data produced by DDAS, compute task-specific pairwise consistency metrics (text: edit distance; table: TEDS; formula: CDM), and classify each sample into three difficulty tiers based on consistency patterns. Since MinerU2.5 is the target model to be improved, we anchor the difficulty taxonomy on its performance relative to external models:

- **Easy:** MinerU2.5’s output is highly consistent with at least one external model. Model consensus indicates the parsing result is reliable, and any model’s output can serve directly as annotation.
- **Medium:** The two external models agree with each other, but MinerU2.5 differs significantly from both. The external consensus serves as a reliable pseudo-label.
- **Hard:** All three models’ outputs exhibit significant pairwise disagreement, and no reliable annotation can be obtained through model consensus.

These three data categories play different roles in training. Easy data is abundant and reliably annotated, forming the backbone for foundational capability building, but the model has largely mastered these scenarios and their marginal training value is limited. Medium data has the highest training value—it precisely pinpoints MinerU2.5’s capability gaps relative to peers, while the successful parsing by external models proves these samples are learnable, and the external consensus directly provides reliable annotations without further correction. Hard data is critical for capability breakthroughs, but its annotations are unreliable and require subsequent Judge-and-Refine correction or expert annotation

(Section 3.3) before safe use. The respective strengths and constraints of these three categories naturally motivate the annotation pipeline described next.

CMCV thus enables rapid difficulty assessment on massive unlabeled data without human annotation, making large-scale data expansion and iteration feasible. Since Medium data is scarce but most valuable, we prioritize its proportion during DDAS sampling. The optimal ratio among the three categories varies by subtask—formula and table recognition are more sensitive to Hard samples, while text recognition benefits more from Medium samples.

3.3 Annotation Pipeline for Hard Case

CMCV provides reliable automatic annotations for Easy and Medium samples. However, Hard samples—data on which all models fail to reach consensus—would introduce annotation noise that degrades rather than improves model performance if directly used for training. Improving annotation quality for these critical samples without relying on large-scale human annotation is the core challenge in advancing the Data Engine from “filtering” to “refinement.” To address this, we design a two-stage pipeline: an automated Judge-and-Refine correction loop followed by targeted expert annotation for residual failures.

Judge-and-Refine Annotation Pipeline. A natural approach to improving Hard sample annotations is to introduce test-time compute through an iterative judge-then-correct mechanism that lets the model examine and refine its own parsing results. However, naive self-reflection exhibits a systematic bias to accept its own outputs: when asked to check its output, the model tends to affirm the result as correct and overlook existing errors. The root cause lies in the asymmetry of cross-modal mapping—models excel at generating structured sequences from document images but struggle to infer visual appearance from structured sequences. For complex structural mappings such as $\LaTeX$ formulas and HTML tables, the model cannot accurately judge how an output sequence will render visually in implicit space, significantly impairing its ability to detect structural errors.

To break this bottleneck, we incorporate *render-then-verify* into the iterative correction loop: we compile $\LaTeX$ formulas and render HTML tables into images, then feed both the original document image and the rendered image to the model as paired inputs alongside the judge-and-refine prompt. This design offers two advantages. First, it closes the missing mapping from structured text to visual layout, reducing the cross-modal reasoning burden. Second, the error-amplification effect of rendering translates subtle, text-domain structural flaws (e.g. missing alignment symbols, unclosed tags) into salient visual anomalies or layout collapse, making defects readily detectable through visual comparison.

Based on this design, we build a visual-comparison-driven Judge-and-Refine iterative correction pipeline. The pipeline uses Qwen3-VL-235B as the Judge-Refine model—chosen for its strong multi-modal reasoning capability and its independence from the CMCV model pool, which avoids systematic bias in error detection. Multi-round error localization and targeted correction proceed via direct visual comparison between the original document image and the rendered image. After processing through this pipeline, a subset of extremely complex cases still remains beyond automatic correction; these samples are routed to the expert annotation workflow.

Targeted Expert Annotation. For Hard samples that remain beyond automatic correction, we introduce expert human annotation to guarantee final quality. Annotation budget is allocated along two priority axes based on intermediate outputs from Judge-and-Refine:

1. *Correction efficiency:* Samples where the Judge stage has localized errors with high confidence but the Refine stage has failed to correct them receive top priority—annotators need only perform local corrections at identified locations, maximizing annotation throughput.

Table 1: Training configurations for the three-stage progressive strategy. All three stages share the same model architecture and resolution settings; they differ in data source, data scale, and optimization hyperparameters, reflecting the progression from broad coverage (Stage 1) to targeted refinement (Stage 2) to metric-level alignment (Stage 3).

Category	Parameter	Stage 1	Stage 2	Stage 3
Vision	Max Resolution	2048×28×28	2048×28×28	2048×28×28
	#Tokens per Image	64–2048	64–2048	64–2048
Data	Dataset Type	Layout & OCR [†] & Image Analysis	Layout & OCR [†] & Image Analysis	Layout & Text & Formula & Table
	#Samples	65.5M	3.9M (192K human-labeled)	192K
Model	Trainable	All	All	All
	Sequence Length	8192	8192	8192
Training	Batch Size	256	128	512
	ViT Learning Rate	1×10^{-4}	5×10^{-6}	1×10^{-7}
	MLP/LLM Learning Rate	1×10^{-3}	5×10^{-5}	1×10^{-5}
	Epoch	1	1	1

[†]OCR collectively refers to text recognition, formula recognition, and table recognition.

2. *Marginal impact*: Within the above pool, priority is further given to the subtask categories where the current model is weakest (determined by CMCV disagreement patterns), maximizing the marginal contribution of limited annotation budget to overall performance.

Human annotation follows an AI pre-annotation and expert review-and-correction workflow. For pre-annotation, we use Gemini 3 Pro—chosen for its strong multimodal reasoning capability and its independence from the CMCV model pool, thereby avoiding data leakage. Automated QA tools further ensure annotation consistency. Compared to MinerU2.5’s human annotation process [25], annotation targets shift from random sampling to a precisely targeted subset identified through three-stage filtering, significantly improving annotation resource utilization.

The Data Engine produces a stratified dataset: approximately 65.5M Easy and Medium samples, automatically annotated via CMCV, are used for Stage 1 pre-training; 192K expert-annotated Hard samples are used for Stage 2 fine-tuning and Stage 3 GRPO alignment.

4 Progressive Training Strategy

MinerU2.5-Pro inherits MinerU2.5’s [25] 1.2B-parameter decoupled coarse-to-fine architecture (NaViT-675M vision encoder + Qwen2-0.5B language model) without any structural modification. The model is initialized from MinerU2.5’s Stage 0 checkpoint, which provides foundational vision-language alignment and OCR capabilities [25].

From this shared starting point, MinerU2.5-Pro employs a three-stage progressive training strategy that sequentially leverages data at different quality tiers produced by the Data Engine: Stage 1 pre-trains on large-scale CMCV auto-annotated data to build comprehensive foundational capabilities; Stage 2 fine-tunes on high-quality expert-annotated data to strengthen performance on hard scenarios; Stage 3 aligns output format and structural conventions through reinforcement learning. The three stages progress from data scale to data quality, with training configurations summarized in Table 1.

4.1 Stage 1: Document Parsing Pre-training

Training data. The training set consists of Easy and Medium samples produced by the Data Engine, with annotations derived from CMCV multi-model consensus. The data covers four core subtasks totaling approximately 65.5M samples: text recognition (21M), layout analysis (14M), formula recognition (13M), and table recognition (11.5M), plus 6M image analysis samples (charts, text-embedded images, *etc.*). Subtask ratios are adjusted based on their weights in the OmniDocBench overall score and the baseline model’s per-task performance gaps.

Training configuration. All parameters are trainable. The language model uses a learning rate of 1×10^{-3} and the vision encoder uses 1×10^{-4} , with a batch size of 256, and training runs for 1 epoch. Compared to MinerU2.5’s Stage 1 pre-training (6.9M samples/epoch $\times$ 2 epochs) [25], this stage expands data scale by nearly an order of magnitude (6.9M $\rightarrow$ 65.5M), with data quality also systematically improved through DDAS distribution correction and CMCV annotation filtering.

4.2 Stage 2: High-Quality Supervised Fine-Tuning

Stage 1 builds comprehensive foundational capabilities, but performance gaps persist on Hard samples. This stage uses high-quality expert-annotated data for targeted fine-tuning, strengthening hard scenarios while maintaining generalization on regular scenarios through mixed Stage 1 replay data.

Training data. The training set comprises two parts: (1) 192K high-quality Hard samples produced through the expert annotation pipeline; (2) replay data sampled proportionally from the Stage 1 training set to prevent catastrophic forgetting. The mixing ratio (Hard:Replay) is differentiated by subtask: layout analysis 6:1, text recognition 1:50, formula recognition 1:25, table recognition 1:10, and image analysis 1:4. This non-uniform mixing strategy reflects differences in hard sample volume and baseline performance across subtasks—layout analysis has more hard samples and a strong Stage 1 foundation, requiring less replay; text recognition has scarce hard samples and requires more replay to preserve generalization.

Training configuration. Building on the Stage 1 model, we adopt a lower learning rate of 5×10^{-5} with a batch size of 128 for 1 epoch. The reduced learning rate protects foundational capabilities acquired in Stage 1 while fine-tuning decision boundaries on hard scenarios.

4.3 Stage 3: Reinforcement Learning with GRPO

The first two stages optimize content recognition accuracy through supervised learning. However, cross-entropy loss optimizes each token prediction independently and weights all tokens equally, without directly reflecting sequence-level or structure-level evaluation metrics (edit distance, CDM, TEDS, IoU). This stage bridges the gap between training objectives and evaluation metrics through reinforcement learning that directly optimizes task-level metrics.

We adopt Group Relative Policy Optimization (GRPO) [31] for alignment. For each input, G groups of candidate outputs are sampled, rewards are computed directly using task-specific automatic evaluation metrics, and policy updates are guided by within-group relative advantages, eliminating the need for a separate reward model.

Reward design. Reward functions are designed separately for four subtasks, directly adopting the same metrics used in evaluation as reward signals: edit distance for text recognition, CDM for formula recognition, TEDS for table recognition, and category IoU for layout detection. This design directly aligns training optimization objectives with final evaluation metrics.

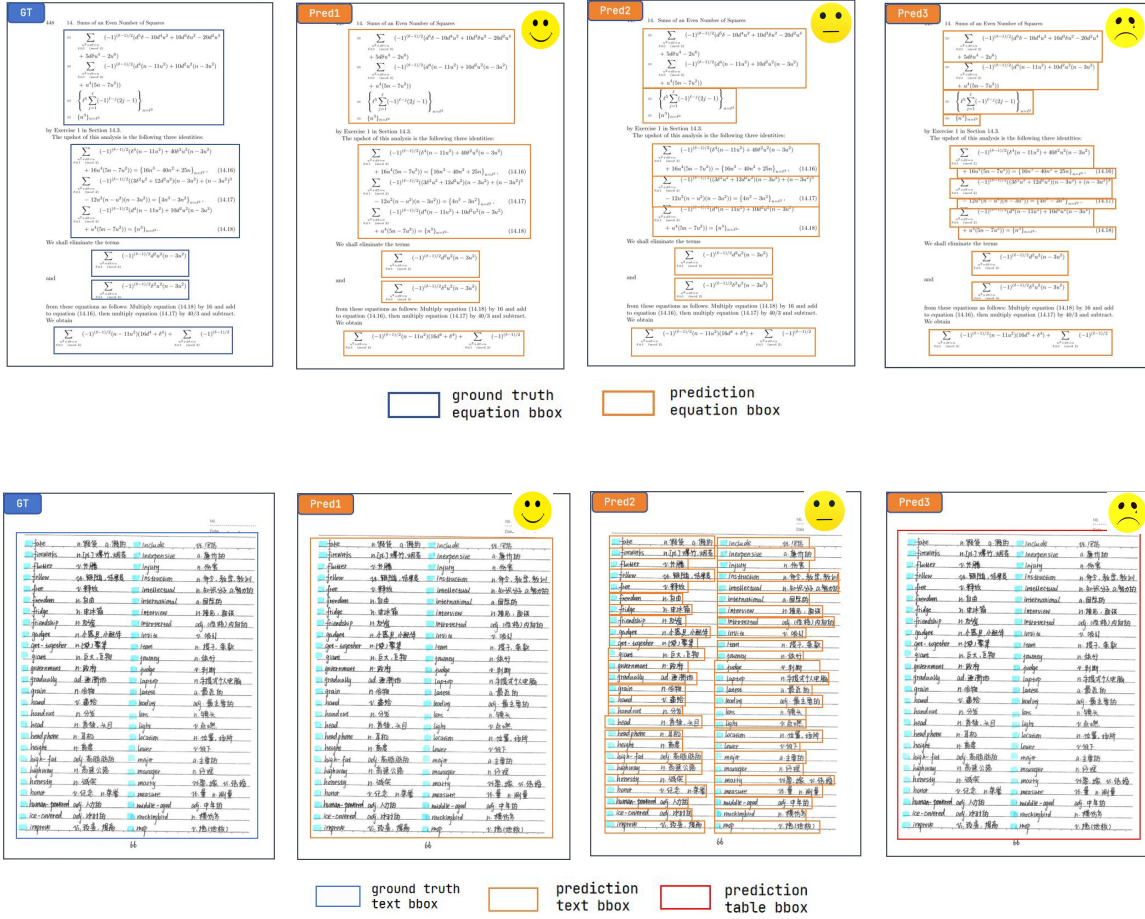


Figure 4: Examples of element-matching bias in OmniDocBench v1.5. Semantically correct predictions receive low scores due to granularity mismatch between predicted and ground-truth segmentation.

Training data. Training data is generated from Stage 2 model rollouts and filtered based on reward distribution: samples with excessively high rewards (model saturated, no effective learning signal) and excessively low rewards (samples too hard or annotations erroneous) are removed, retaining the mid-reward range to maximize effective policy gradient signal. All training data comes from the high-quality expert-annotated set to ensure reward signal reliability.

Training configuration. Building on the Stage 2 model, we adopt a learning rate of 1×10^{-5} with a batch size of 512 for 1 epoch, sampling $G = 16$ rollouts per sample. Following DAPO [46], we apply clip-higher to stabilize advantage estimation and dynamic sampling to discard zero-variance rollout groups.

5 OmniDocBench v1.6

5.1 Motivation

As leading document parsing models converge on OmniDocBench v1.5, two fundamental issues limit its effectiveness as a benchmark:

Matching strategy bias. v1.5 employs fixed-granularity one-to-one element matching, which silently penalizes systems whose output segmentation differs from the ground truth—even when the parsed content is entirely correct. As illustrated in Figure 4, consider a multi-line formula annotated as a single block spanning k lines: if a model produces the identical $\text{\LaTeX}$ but segments it into $k-1$ or k separate blocks, scores drop sharply from full marks to near zero despite semantically perfect output. A similar issue affects dense text: a region annotated as one block may be predicted line-by-line or even recognized as a table; in the latter case v1.5 assigns zero credit because no text element remains to match. These granularity-dependent scoring artifacts make cross-system comparisons unreliable.

Insufficient hard sample coverage. Through the large-scale difficulty stratification provided by our Data Engine (Section 3), we find that samples labeled as Hard are virtually absent from the v1.5 evaluation set. The benchmark predominantly measures performance on low-to-medium difficulty documents, causing top models to cluster tightly with diminishing discriminative power.

To address these issues, we upgrade OmniDocBench to v1.6: we propose Multi-Granularity Adaptive Matching (MGAM) to eliminate matching bias (Section 5.2), and expand the evaluation set with a dedicated Hard subset (Section 5.3).

5.2 Multi-Granularity Adaptive Matching

We propose **Multi-Granularity Adaptive Matching (MGAM)**, which eliminates matching bias through adaptive granularity adjustment on the prediction side. The core principle is to keep the ground truth unchanged and search for the optimal segmentation granularity only on the prediction side. Given a set of ground truth elements $\mathcal{G} = \{g_1, g_2, \dots, g_m\}$ and prediction elements $\mathcal{P} = \{p_1, p_2, \dots, p_n\}$, MGAM generates candidate matching solutions through three stages and selects the global optimum:

Stage 1: Direct Bipartite Matching. We directly solve optimal bipartite matching between $\mathcal{P}$ and $\mathcal{G}$ at original granularity. Using the cost matrix $C_{ij} = 1 - \text{sim}(p_i, g_j)$ as input, the Hungarian algorithm solves for the minimum-cost matching $\mathcal{M}_1^* = \arg \min_{\mathcal{M}} \sum_{(i,j) \in \mathcal{M}} C_{ij}$, yielding the first candidate matching and its aggregate score S_1 .

Stage 2: Prediction Splitting + Bipartite Matching. Each prediction element p_i is split at $\text{\LaTeX}$ line-break delimiters (e.g. `\\`, `\newline`, and equivalent symbols) to produce a fine-grained prediction set $\mathcal{P}' = \{p'_1, p'_2, \dots, p'_{n'}\}$ ($n' \geq n$). Prediction elements without splittable delimiters remain unchanged. Bipartite matching is re-solved on $\mathcal{P}'$ and $\mathcal{G}$, yielding candidate matching $\mathcal{M}_2^*$ and aggregate score S_2 .

Stage 3: Partition Enumeration + Bipartite Matching. Stage 2 splitting may be too fine—annotation granularity is not necessarily per-line but may be any intermediate granularity between 1 and k lines. To cover all possible merging schemes, we enumerate all valid ordered partitions of consecutive subsequences of $\mathcal{P}'$. Specifically, for n' fine-grained prediction elements, there are $n' - 1$ gaps between adjacent elements, and each gap can be either “split” or “merged,” producing $2^{n'-1}$ partition schemes. Each partition $\pi = (B_1, B_2, \dots, B_K)$ divides $\mathcal{P}'$ into K consecutive blocks, where the k -th block is

$$B_k = \bigoplus_{t=l_k}^{r_k} p'_t \quad (1)$$

with $\bigoplus$ denoting concatenation in original order. For each partition, we perform bipartite matching between the merged block set $\{B_1, \dots, B_K\}$ and $\mathcal{G}$, selecting the partition with the best matching score as candidate matching $\mathcal{M}_3^*$ and aggregate score S_3 .

Global Optimum Selection. The final matching is the one with the best aggregate score among the three stages: $\mathcal{M}^* = \arg \max_{k \in \{1,2,3\}} S_k$, and the task-specific metric (e.g. CDM for formulas, edit distance for text) is computed based on $\mathcal{M}^*$.

Dense text matching. The granularity mismatch problem is not limited to formulas. For dense text regions, prediction and annotation sides similarly differ in whether multiple text segments are merged into one large text box or split into multiple small ones. We reuse the MGAM algorithm for text elements with edit distance as the similarity metric. Additionally, if a model recognizes text in a region as a table (not uncommon for dense structured text), we convert the table back to plain text and include it in the same matching pipeline, avoiding unfair penalties due to format preference differences.

With MGAM, the evaluation becomes neutral to output granularity and format preferences, removing a systematic source of scoring variance across systems.

5.3 Hard Subset and Three-Tier Evaluation Protocol

To fill the coverage gap for hard scenarios, we construct a Hard subset of 296 pages selected from the pool of data labeled as Hard during Data Engine difficulty stratification. Samples are chosen to cover the most challenging scenario categories in document parsing, including complex nested tables, dense mathematical formula layouts, and unconventional layout structures. All Hard subset samples are excluded from every training stage of MinerU2.5-Pro (including Judge-and-Refine training data) and are annotated by professional teams with inter-annotator cross-validation to ensure ground truth quality.

OmniDocBench v1.6 establishes a Base/Hard/Full three-tier evaluation protocol:

- **Base** (1,355 pages): retains the original v1.5 evaluation set to maintain comparability with historical results.
- **Hard** (296 pages): the newly added hard sample subset, providing more sensitive measurement where top models saturate on standard evaluations.
- **Full** (1,651 pages): the complete union of both, providing comprehensive performance assessment.

6 Experiments

This section evaluates MinerU2.5-Pro against both leading general-purpose VLMs and current SOTA document-parsing-specific models [5, 43, 33, 11, 8, 32, 41, 40, 25]. All competing models are re-evaluated under a unified environment using the same evaluation code.

6.1 Evaluation Setup

OmniDocBench v1.6. We evaluate on OmniDocBench v1.6 using the Base/Hard/Full three-tier protocol described in Section 5.3. The overall score follows the same formula as MinerU2.5 [25], averaging text (edit distance), table (TEDS), and formula (CDM) metrics. We additionally report sub-metrics: Text Edit↓, Formula CDM↑, Table TEDS↑, Table TEDS-S↑, Read Order Edit↓.

Element-specific evaluation. To more accurately measure content recognition capability (excluding the confounding effect of layout detection errors), we crop document images based on ground truth layout boxes and separately evaluate text recognition, formula recognition, and table recognition as individual modules.

Table 2: Performance comparison of document parsing methods on OmniDocBench v1.6 Full across text, formula, table, and reading order extraction tasks.

Model Type	Methods	Param	Overall $\uparrow$	Text $\text{Edit}\downarrow$	Formula $\text{CDM}\uparrow$	Table $\text{TEDS}\uparrow$	Table $\text{TEDS-S}\uparrow$	Read Order $\text{Edit}\downarrow$
Specialized VLMs	MinerU2.5-Pro	1.2B	95.69	0.036	97.29	93.42	95.92	0.120
	GLM-OCR [11]	0.9B	95.15	0.044	96.99	92.83	95.39	0.133
	PaddleOCR-VL-1.5 [8]	0.9B	94.87	0.038	96.69	91.67	94.37	0.130
	PaddleOCR-VL [6]	0.9B	94.11	0.040	95.70	90.65	93.74	0.135
	Youtu-Parsing [45]	2.5B	93.68	0.044	93.45	92.02	95.00	0.116
	Logics-Parsing-v2 [4]	4B	93.27	0.041	95.47	88.42	91.98	0.137
	FireRed-OCR [41]	2B	93.20	0.037	95.27	88.04	91.06	0.131
	MinerU2.5 [25]	1.2B	92.98	0.045	95.59	87.88	91.47	0.130
	OpenDoc-0.1B [10]	0.1B	90.64	0.049	92.93	83.88	87.45	0.140
	dots.ocr [18]	3B	90.50	0.048	89.12	87.18	90.58	0.138
	DeepSeek-OCR 2 [40]	3B	90.17	0.050	91.59	83.89	87.75	0.144
	HunyuanOCR [32]	1B	89.87	0.089	87.44	91.01	93.23	0.171
	Dolphin-v2 [12]	3B	89.34	0.069	90.53	84.40	87.44	0.150
	OCRVerse [52]	4B	88.44	0.063	89.14	82.44	86.27	0.163
	MonkeyOCR-pro-3B [19]	3B	88.43	0.074	88.33	84.35	88.62	0.189
General VLMs	Ovis2.6-30B-A3B [22, 23]	30B	93.62	0.035	94.93	89.44	92.40	0.135
	Gemini 3 Pro	–	92.85	0.064	95.83	89.15	92.96	0.165
	Gemini 3 Flash	–	92.58	0.066	95.03	89.29	93.51	0.173
	Qwen3-VL-235B [43]	235B	89.78	0.063	92.53	83.07	86.75	0.166
	GPT-5.2	–	86.52	0.114	88.00	82.95	87.93	0.193
	InternVL3.5-241B [38]	241B	83.61	0.130	89.52	74.35	79.78	0.215

Table 3: Training stage ablation on OmniDocBench v1.6.

Stage	Base	Hard	Full	ΔFull	Text $\downarrow$	CDM $\uparrow$	TEDS $\uparrow$
MinerU2.5 (baseline)	93.23	91.65	92.98	—	0.045	95.59	87.88
Stage 1: Large-Scale SFT	94.54	93.10	94.29	+1.31	0.039	96.40	90.37
+ Stage 2: Hard-Sample SFT	95.60	93.84	95.25	+0.96	0.036	96.48	92.87
+ Stage 3: GRPO	96.12	94.08	95.69	+0.45	0.036	97.29	93.42

6.2 End-to-End Document Parsing

As shown in Table 2, MinerU2.5-Pro ranks first on Full with 95.69, improving over the same-architecture MinerU2.5 baseline (92.98) by 2.71 points—confirming that all gains are data-driven. On the Base subset Table 7, the top three models (GLM-OCR 96.19, MinerU2.5-Pro 96.12, PaddleOCR-VL-1.5 95.72) are within 0.5 points, indicating near-saturation on standard scenarios. On the Hard subset Table 8, MinerU2.5-Pro leads at 94.08, exceeding both GLM-OCR and PaddleOCR-VL-1.5 (both at 92.01) by 2.07 points, demonstrating the Data Engine’s advantage in hard scenario robustness and validating the Hard subset’s discriminative power.

Across sub-metrics, MinerU2.5-Pro achieves the best scores in formula recognition (CDM 97.29), table recognition (TEDS 93.42, TEDS-S 95.92), and reading order (0.120). Notably, Gemini 3 Pro/Flash benefit substantially from the corrected matching in OmniDocBench v1.6 (Full 92.85/92.58), narrowing the gap with specialized models, yet specialized models at only 0.9B–1.2B parameters maintain an overall lead.

Training stage ablation. Table 3 reports the incremental contribution of each training stage.

Stage 1 (large-scale SFT) contributes the largest single-stage gain (+1.31), indicating that the Data Engine’s optimization of data coverage and annotation quality is the primary driver of performance improvement. Stage 2 (hard sample fine-tuning) adds +0.96, with the most notable contribution in table recognition (TEDS 90.37 $\rightarrow$ 92.87, +2.50). Stage 3 (GRPO) contributes +0.45, primarily reflected in formula CDM improvement (96.48 $\rightarrow$ 97.29, +0.81), driven by reinforcement learning’s direct optimization of task-level metrics. The cumulative improvement on the Hard subset (91.65 $\rightarrow$ 94.08, +2.43) is comparable to the Base subset (93.23 $\rightarrow$ 96.12, +2.89), indicating that the progressive training

Table 4: Text recognition (Edit Distance↓) on OmniDocBench v1.6.

Model	Type	Base↓	Hard↓	Full↓
MinerU2.5-Pro	Decoupled	0.015	0.048	0.019
Qwen3.5-397B [33]	General	<u>0.016</u>	0.052	<u>0.020</u>
GLM-OCR [11]	Decoupled	0.016	0.053	0.021
Qwen3-VL-235B [43]	General	0.017	<u>0.049</u>	0.021
PaddleOCR-VL-1.5 [8]	Decoupled	0.018	0.056	0.022
MinerU2.5 [25]	Decoupled	0.023	0.066	0.028
PaddleOCR-VL [6]	Decoupled	0.019	0.057	0.023
DeepSeek-OCR 2 [40]	End-to-End	0.057	0.130	0.066
FireRed-OCR [41]	End-to-End	0.135	0.176	0.140

Table 5: Formula recognition (CDM↑) across multiple benchmarks. CPE, HWE, SCE, and SPE are the Complex Printed, Handwritten, Screen-Captured, and Simple Printed Expression subsets from UniMERNet [34], respectively.

Model	OmniDoc		Public					Inhouse	
	Base	Hard	CPE	HWE	SCE	SPE	LaTeX-80M	Chinese	Fuzzy
MinerU2.5-Pro	99.20	98.79	98.97	95.38	97.04	99.44	97.23	95.28	94.90
PaddleOCR-VL-1.5 [8]	<u>98.76</u>	97.22	98.84	92.27	94.95	99.27	92.77	94.06	89.73
GLM-OCR [11]	98.75	98.28	96.74	95.10	97.77	98.42	95.39	94.35	93.75
PaddleOCR-VL [6]	98.72	97.64	<u>98.93</u>	94.45	95.88	99.30	93.67	94.35	91.56
Qwen3.5-397B [33]	98.19	97.25	98.32	97.59	95.87	<u>99.41</u>	95.17	78.24	90.53
Qwen3-VL-235B [43]	97.72	98.13	97.47	94.23	96.21	98.46	95.33	92.69	93.59
MinerU2.5 [25]	97.25	<u>98.67</u>	97.79	94.42	96.65	98.57	<u>96.23</u>	95.50	94.92
FireRed-OCR [41]	96.71	94.54	94.35	85.42	89.94	96.75	83.41	87.94	87.77
DeepSeek-OCR 2 [40]	95.95	93.39	91.97	81.67	77.19	95.51	72.04	87.82	85.13

strategy achieves balanced capability improvement across both hard and standard scenarios.

6.3 Element-Specific Parsing

Layout detection accuracy in end-to-end evaluation cascades into content recognition scores, and differences in output granularity and segmentation strategies across models prevent precise matching of a small number of elements. To more fairly evaluate pure content recognition capability, we crop document images based on ground truth layout boxes and test text, formula, and table recognition as individual modules. Note that end-to-end models do not receive element category priors in this setting, which may partially explain their larger performance gap compared to decoupled two-stage models.

Text recognition. As shown in Table 4, MinerU2.5-Pro ranks first on Full with an edit distance of 0.019, a 30.5% reduction from the MinerU2.5 baseline (0.028). Hundred-billion-scale general VLMs (Qwen3.5-397B, Qwen3-VL-235B) demonstrate competitive text recognition performance comparable to specialized models, while end-to-end models (DeepSeek-OCR 2, FireRed-OCR) show significant degradation without category priors.

Formula recognition. Table 5 reports CDM scores across 9 benchmarks. MinerU2.5-Pro achieves the best score in 5 dimensions and the second-highest in the remaining four. Specifically, it falls short on HWE (handwritten formulas) against Qwen3.5-397B (95.38 *vs.* 97.59) and on SCE against GLM-OCR (97.04 *vs.* 97.77), while only trailing slightly behind MinerU2.5 on Chinese and Fuzzy subsets. On OmniDocBench Base, CDM reaches 99.20 (out of 100), approaching the performance ceiling for formula

Table 6: Table recognition (TEDS & TEDS-S $\uparrow$) across multiple benchmarks.

Model	Type	OmniDoc Base		OmniDoc Hard		CCOCR		OCRBv2		Inhouse		Overall	
		TEDS	TEDS-S	TEDS	TEDS-S	TEDS	TEDS-S	TEDS	TEDS-S	TEDS	TEDS-S	TEDS	TEDS-S
MinerU2.5-Pro	Decoup.	<u>95.67</u>	<u>97.42</u>	92.46	94.67	88.49	<u>91.90</u>	93.56	96.27	82.70	89.65	91.10	94.48
GLM-OCR [11]	Decoup.	96.14	97.60	<u>90.49</u>	<u>93.47</u>	89.17	92.58	91.19	94.44	<u>79.41</u>	<u>87.65</u>	<u>89.71</u>	<u>93.52</u>
Gemini 3 Pro	General	94.42	97.37	88.16	91.34	86.47	90.10	91.73	94.96	75.91	85.26	88.21	92.65
PaddleOCR-VL-1.5 [8]	Decoup.	93.85	95.57	88.28	91.79	76.34	81.38	82.64	86.91	72.66	81.93	82.91	87.60
Qwen3.5-397B [33]	General	93.76	96.27	89.67	93.19	<u>88.87</u>	91.89	88.11	91.21	77.49	85.53	87.60	91.57
Qwen3-VL-235B [43]	General	92.92	95.48	87.55	91.80	87.38	91.22	88.81	92.94	74.39	83.94	86.64	91.44
MinerU2.5 [25]	Decoup.	92.87	95.33	88.28	91.80	84.35	88.25	<u>92.32</u>	<u>95.31</u>	76.95	85.98	87.94	92.17
PaddleOCR-VL [6]	Decoup.	92.31	94.64	89.21	92.02	81.54	85.80	82.29	85.79	74.88	82.92	83.67	87.85
FireRed-OCR [41]	E2E	88.12	90.29	86.11	89.58	82.24	86.62	86.93	90.34	69.57	79.02	83.02	87.47
DeepSeek-OCR 2 [40]	E2E	79.27	82.46	75.07	80.06	66.98	72.74	84.38	88.70	57.70	69.35	74.59	80.35

recognition. While Qwen3.5-397B excels on handwritten formulas, it reveals a notable weakness on Chinese formulas (Chinese 78.24).

Table recognition. As shown in Table 6, MinerU2.5-Pro ranks first in both Overall TEDS (91.10) and TEDS-S (94.48), improving over MinerU2.5 by 3.16 and 2.31 percentage points, respectively. The advantage is most pronounced on the Hard subset (TEDS 92.46 *vs.* MinerU2.5’s 88.28, +4.18), indicating that the Data Engine’s hard sample mining and expert annotation contribute most to table recognition. GLM-OCR is slightly better on OmniDocBench Base (96.14) and CCOCR (89.17) but is less stable than MinerU2.5-Pro across benchmarks. PaddleOCR-VL-1.5 shows notable performance drops on CCOCR (TEDS 76.34) and Inhouse (TEDS 72.66), suggesting limited table recognition generalization.

7 Conclusion

We present MinerU2.5-Pro, which improves the OmniDocBench v1.6 overall score from 92.98 to 95.69 solely through systematic data engineering while keeping the 1.2B-parameter model architecture completely fixed, surpassing all existing methods. This result demonstrates that at the current stage where architectures are maturing, co-optimizing training data coverage, informativeness, and annotation accuracy yields greater performance gains than architectural improvements alone. To this end, we contribute a Data Engine that expands training data from under 10M to 65.5M pages while systematically improving annotation quality, a three-stage progressive training strategy matched to data quality tiers, and the OmniDocBench v1.6 three-tier evaluation protocol that corrects evaluation biases. These tools and methodologies provide the community with a performance improvement pathway that is orthogonal to and complementary with architectural innovation.

Limitations and Future Directions

Fundamental challenges in evaluation. OmniDocBench v1.6 improves scoring fairness through corrected matching strategies, but the element-matching paradigm itself has inherent limitations. The ambiguity is twofold: at the format level, the same content can be expressed in multiple equivalent notations (*e.g.* HTML *vs.* Markdown for tables, different $\text{\LaTeX}$ commands for the same formula); at the structural level, the same visual layout can be legitimately represented with different element types—for instance, a bilingual word list with aligned Chinese and English columns is equally valid as line-by-line text pairs or as a two-column table, and even human annotators may disagree on which representation is “correct.” Developing semantic-equivalence-aware evaluation methods that account for both format and structural ambiguity remains an open problem.

Evaluation coverage and domain adaptation. OmniDocBench v1.6 aims to cover mainstream application scenarios; for vertical domains with higher precision requirements (*e.g.* finance, legal, medical),

constructing domain-specific evaluation sets is a necessary complement. Furthermore, as model capabilities approach human-level performance, ensuring the precision of evaluation set annotations themselves becomes an increasingly pressing challenge.

From parsing accuracy to structural understanding. This work focuses on content accuracy in document parsing. However, for downstream applications, structural relationships within documents—such as hierarchical relationships between headings and body text, semantic bindings between figures/tables and referring text, and cross-page content continuity—are equally critical for document retrieval and downstream semantic understanding. Advancing parsing from “content extraction” to “structured semantic understanding” represents a natural next step for document parsing research.

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Appendix

A Prompt Design and Task Examples

This section provides the prompt formats, output specifications, and representative examples for each task supported by MinerU2.5-Pro. All tasks share a unified prompt interface: a single `<image>` token followed by a plain-text task suffix, requiring no few-shot examples or structured metadata.

Task Prompt Convention. The five task suffixes and their output formats are summarized below:

- **Layout Detection** (§A.1) — localizes content regions and outputs bounding boxes with category labels and rotation flags.
- **Text Recognition** (§A.2) — transcribes cropped text regions into plain text.
- **Formula Recognition** (§A.3) — converts cropped formula regions into $\text{\LaTeX}$ markup.
- **Table Recognition** (§A.4) — serializes cropped tables into an OTSL-based token sequence with cell content, subsequently converted to HTML.
- **Image Analysis** (§A.5) — classifies image regions and extracts captions and embedded content.

A.1 Layout Detection

Layout Detection serves as the entry point of the document parsing pipeline, responsible for localizing all content regions on a page and assigning each a semantic category. The model takes a downsampled page image and produces a sequence of structured region descriptors.

Prompt.

`<image>\nLayout Detection:`

Output Format. The output is a newline-delimited sequence of region descriptors, where each region follows the format:

```
<|box_start|>x1 y1 x2 y2<|box_end|><|ref_start|>category<|ref_end|><|rotate_dir|>
```

Here, `x1 y1 x2 y2` are the normalized bounding box coordinates (scaled to a `[0, 999]` grid), `category` is the semantic label of the region (e.g., `title`, `text`, `header`, `footer`, `table`, `figure`, `formula`), and `<|rotate_dir|>` indicates the text orientation (`<|rotate_up|>` for standard upright text, with other directions for rotated content). Regions are emitted in natural reading order (top-to-bottom, left-to-right for left-to-right scripts).

Example. Given a document title page, the model outputs:

```
<|box_start|>705 112 899 146<|box_end|><|ref_start|>header<|ref_end|><|rotate_up|>
<|box_start|>030 343 132 397<|box_end|><|ref_start|>title<|ref_end|><|rotate_up|>
<|box_start|>212 330 491 382<|box_end|><|ref_start|>title<|ref_end|><|rotate_up|>
<|box_start|>214 389 767 441<|box_end|><|ref_start|>title<|ref_end|><|rotate_up|>
<|box_start|>219 494 359 523<|box_end|><|ref_start|>text<|ref_end|><|rotate_up|>
<|box_start|>654 940 907 975<|box_end|><|ref_start|>footer<|ref_end|><|rotate_up|>
```

This output identifies six regions—one header, three title blocks, one text block, and one footer—each with precise spatial coordinates and upright orientation. Additional examples are provided in [Figure 5](#).

A.2 Text Recognition

Text Recognition transcribes cropped text regions into plain text. Each region is an original-resolution crop produced by Stage 1 Layout Detection.

Prompt.

<image>\nText Recognition:

Output Format. The output is a plain-text string corresponding to the content of the cropped text region. No special tokens or markup are used—the model generates raw text as-is, including whitespace, punctuation, and any inline symbols present in the source image. Additional examples are provided in [Figure 6](#).

A.3 Formula Recognition

Formula Recognition converts cropped formula regions into $\LaTeX$ markup. The model supports both inline and display-style formulas, as well as multi-line equation environments.

Prompt.

<image>\nFormula Recognition:

Output Format. The output is a $\LaTeX$ math string. Display-style (block) formulas are wrapped in $\left[\dots \right]$ delimiters. Equation numbers, when present in the source image, are preserved via $\tag{\dots}$. The model generates standard $\LaTeX$ math commands and environments (e.g., $\frac{}{} \mathrel{\mathop{\kern 0pt}{\rm cm}}$), ensuring the output is directly compilable.

Example: Single Display Formula. Given a cropped equation region, the model outputs:

$$\left[L = 0.004 \ln \left(2D/d \right) \quad \left(\mu \mathrel{\mathop{\kern 0pt}{\rm H}} / \mathrel{\mathop{\kern 0pt}{\rm cm}} \right) \tag{4-2-10} \right]$$

The equation number (4-2-10) from the original document is captured via $\tag{\dots}$.

Multi-line Formulas. Multi-line equations are handled through the collaboration of Layout Detection and Formula Recognition. Layout Detection first identifies an `equation_block` region encompassing the entire multi-line group, within which individual single-line formulas are separately localized. Each line is then independently cropped and recognized by Formula Recognition. The final multi-line output is produced by concatenating the individual $\LaTeX$ results in reading order, faithfully reproducing the original equation group without requiring the model to generate multi-line environments in a single pass.

Additional examples are provided in [Figure 7](#).

A.4 Table Recognition

Table Recognition converts cropped table regions into a structured token sequence based on OTSL (Optimized Table Structure Language). Cell content is transcribed as plain text, with inline formulas in $\LaTeX$ when present. The OTSL output is subsequently converted to HTML for downstream consumption.

Prompt.

<image>\nTable Recognition:



Figure 6: Text Recognition examples across Chinese, English, and mixed-language text regions.

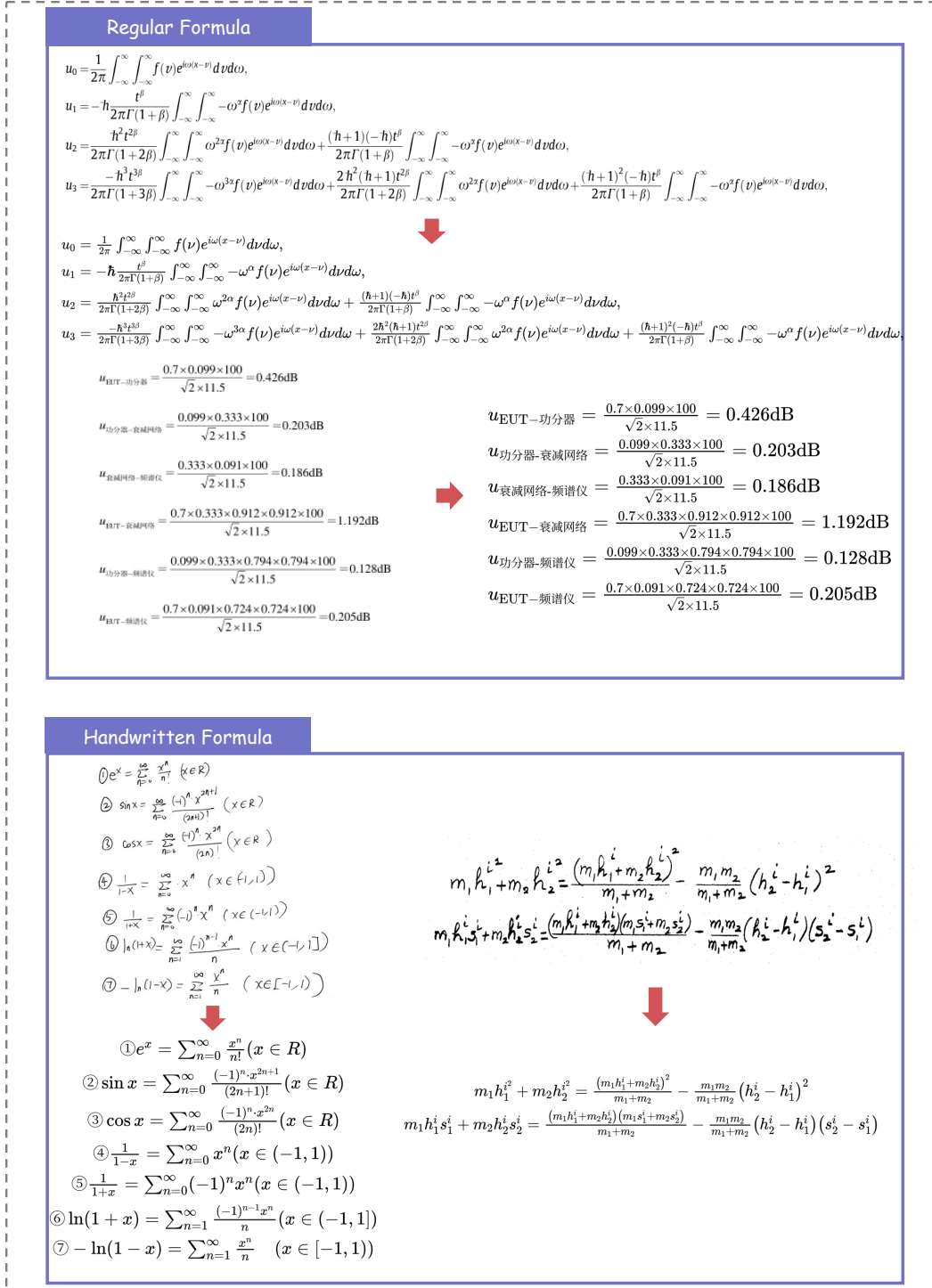


Figure 7: Formula Recognition examples including single-line display formulas and complex multi-line equations.

Output Format. The output is a flat token sequence representing the table structure row by row. Each cell is delimited by `<fcel>`, and rows are separated by `<n1>`. Cell content may contain plain text, $\LaTeX$ inline math (`\( . . . \)`), or a mixture of both. The OTSL representation is compact and unambiguous, supporting regular grids as well as cells with complex content. After generation, the OTSL sequence is programmatically converted to HTML for rendering and downstream integration.

Example. Given a cropped table with two rows (a header row of time values and a data row of concentration values), the model outputs:

```
<fcel>\( \frac{t}{t_{\min}} \)<fcel>3<fcel>5<fcel>7<fcel>10<fcel>15<fcel>21<fcel>25<n1>
<fcel>\( \frac{1}{c} \{ \mathrm{\backslash;{mol}} \} ^{-1} \cdot \{ \mathrm{{dm}} \} ^{3} \)<fcel>135.1
<fcel>157.7<fcel>181.8<fcel>215.5<fcel>275.5<fcel>347.2<fcel>393.2<n1>
```

Each `<fcel>` token introduces a cell, and `<n1>` marks the end of a row. The first column contains $\LaTeX$ -formatted headers with units, while the remaining columns hold numeric values.

Additional examples are provided in [Figure 8](#).

A.5 Image-Aware Parsing

Image Analysis classifies cropped image regions and extracts their embedded content. Unlike other recognition tasks that target a single modality, Image Analysis first determines the semantic type of the image and then extracts structured content accordingly—text, formulas, tables, or a combination thereof.

Prompt.

`<image>\nImage Analysis:`

Output Format. The output consists of four structured fields delimited by special tokens:

```
<|class_start|>class<|class_end|>
<|sub_class_start|>sub_class<|sub_class_end|>
<|caption_start|>caption<|caption_end|>
<|content_start|>content<|content_end|>
```

Here, `class` is the primary image category (e.g., `pure_formula`, `natural_image`, `chart`), `sub_class` provides a finer-grained label, `caption` captures any associated caption text (left empty if absent), and `content` contains the extracted textual or structured content from within the image.

Example. Given a cropped figure region containing a standalone formula, the model outputs:

```
<|class_start|>pure_formula<|class_end|>
<|sub_class_start|>pure_formula<|sub_class_end|>
<|caption_start|><|caption_end|>
<|content_start|>p + q = 1<|content_end|>
```

The image is classified as `pure_formula` with no caption, and the formula content is directly extracted. Additional examples are provided in [Figure 9](#).

B Extended Parsing Capabilities

Beyond improvements in recognition accuracy, MinerU2.5-Pro extends the parsing capabilities of MinerU2.5 in several practical dimensions. These features target real-world deployment scenarios

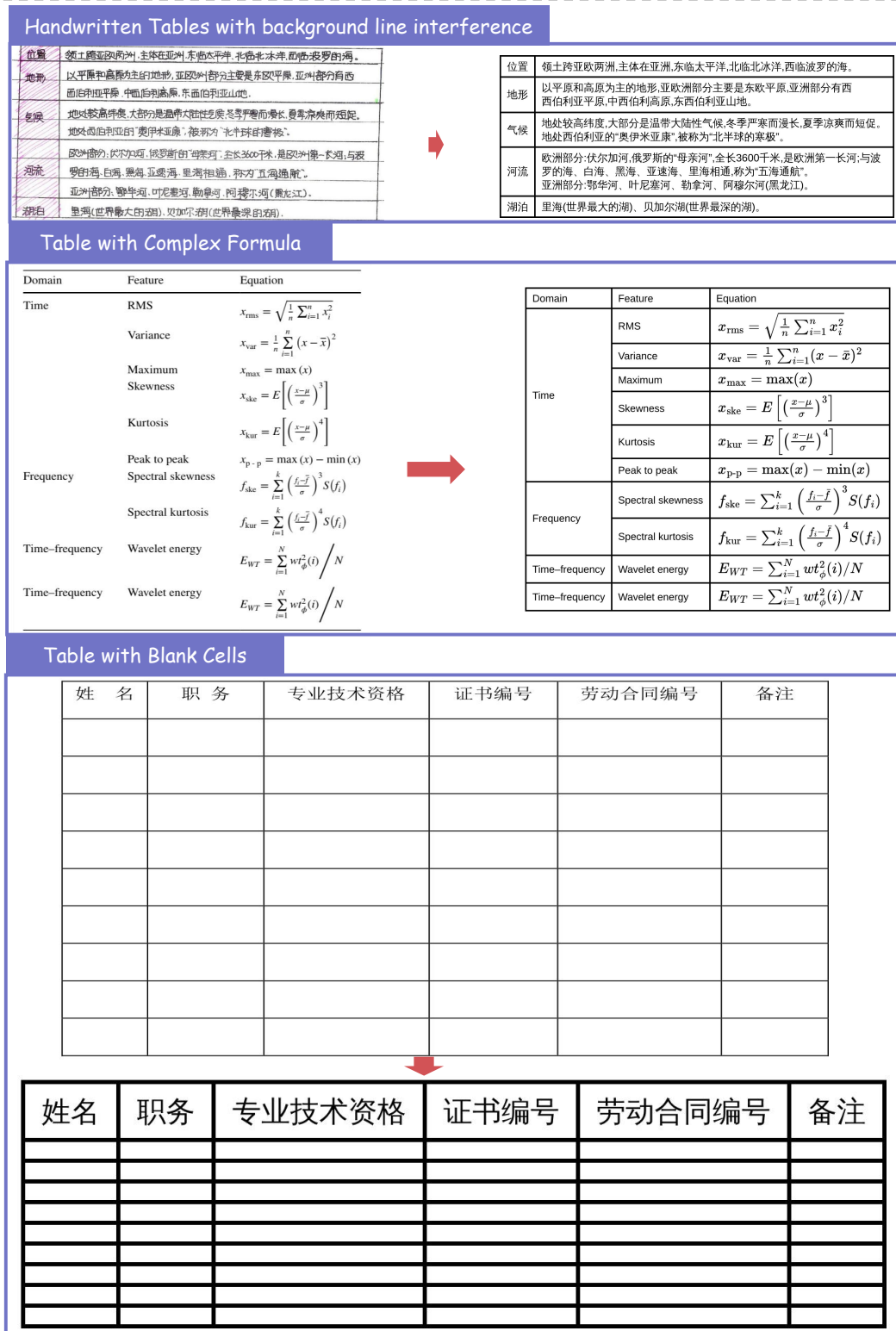


Figure 8: Table Recognition examples showing OTSL token output and the corresponding rendered HTML for tables with varying complexity.

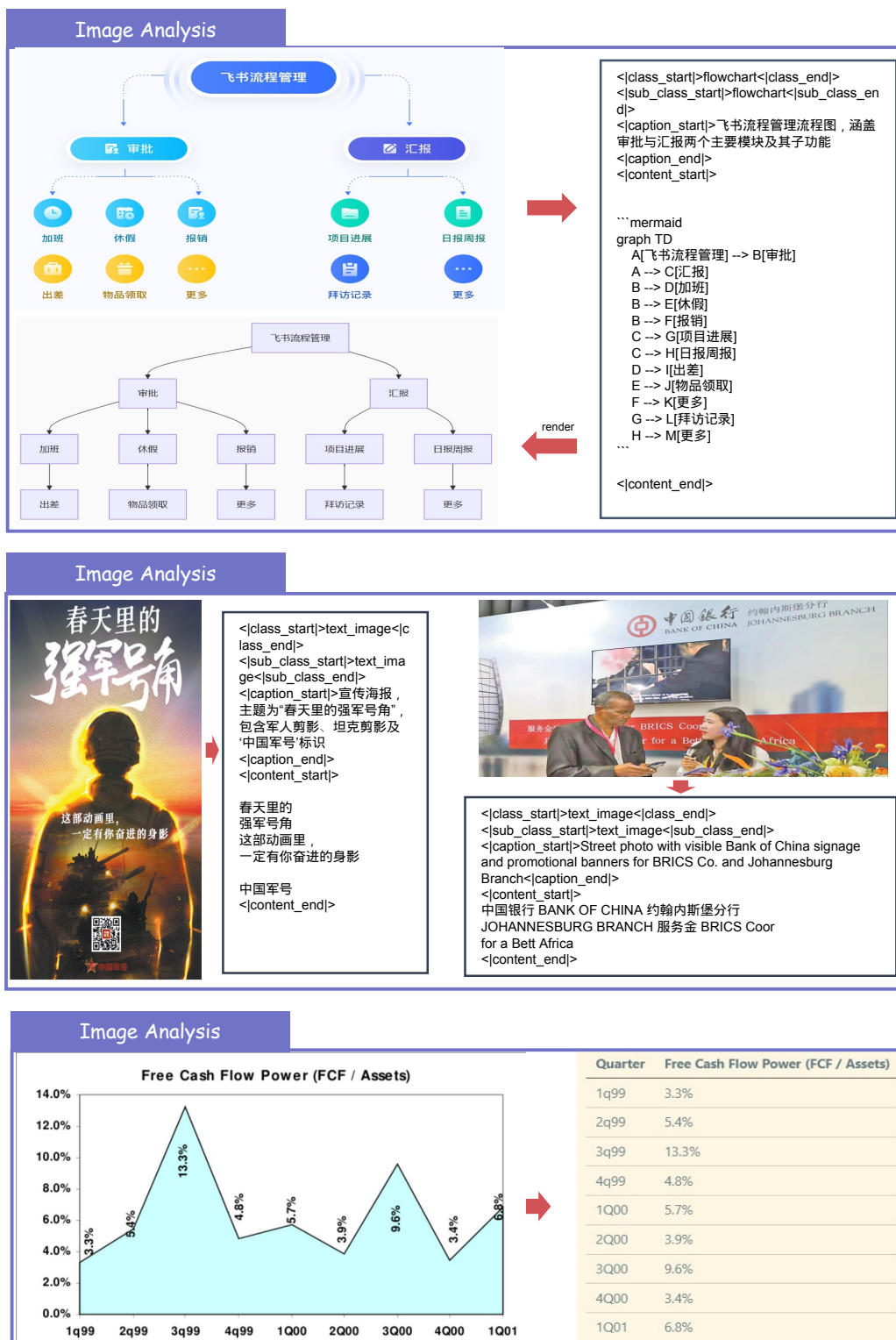


Figure 9: Image-aware parsing examples. The model classifies image regions into fine-grained subtypes and extracts structured content accordingly.

where documents are multi-page, richly illustrated, and structurally complex. While they do not affect OmniDocBench scores (which focus on single-page content recognition), they substantially improve end-to-end parsing completeness and usability.

Image-aware parsing. MinerU2.5 crops all image regions without further processing, discarding potentially valuable information such as chart data, embedded text, and diagram content. MinerU2.5-Pro introduces image-aware parsing (§A.5) that first classifies each image region into fine-grained subtypes (chart, text image, table-like image, general image) and then applies differentiated extraction strategies: charts are parsed into structured tables, text images undergo OCR, and table-like images are recognized as tables. This framework is readily extensible to additional image types; however, we have not yet applied Data Engine optimization to image analysis data in this release, leaving significant room for future improvement.

Truncated paragraph merging. Layout Detection tends to segment each spatially distinct text block as an independent region, which can split semantically continuous paragraphs into multiple fragments. Common causes include column boundaries in multi-column layouts, figures or tables interrupting a paragraph, and unusually wide line spacing. To address this, MinerU2.5-Pro performs *truncated paragraph merging* as part of the Layout Detection task. Since Layout Detection already establishes reading order, and truncation necessarily occurs between consecutive regions in that order, the problem reduces to a binary classification at each adjacent-region boundary: *merge* or *no merge*. This binary label is integrated directly into the layout output sequence, allowing truncated paragraphs to be reassembled during final Markdown rendering without affecting downstream recognition tasks. The merging process is illustrated in Figure 10.

To construct training data for this capability, we annotate merge decisions on top of existing layout ground truth. For each pair of adjacent text or `list_item` regions, we first apply rule-based filtering using sentence length, leading numbering patterns, and terminal punctuation to eliminate obvious non-merge cases. For the remaining candidates, we highlight the two regions in red and green on the page image and query Gemini 3 Flash with both the annotated image and the text content of each region, asking it to judge whether merging is appropriate based on layout context and textual coherence. To reduce API cost, only the first and last sentences are provided for long paragraphs.

Cross-page table merging. When a table is split across a page break, MinerU2.5-Pro automatically detects and merges the fragments. The system first applies rule-based heuristics to identify candidate pairs: if the last table on a page and the first table on the next page share compatible column counts and structural patterns, they are flagged for merging. For flagged pairs, the model receives the last row(s) of the upper table and the first data row(s) of the lower table as a structured text prompt:

Please merge the next two tables.

```
## Table 1 (Previous Page - Last Table)
**Last Row(s) Data:**
[[{content of table 1}]]
```

```
## Table 2 (Current Page - First Table)
**First Data Row(s):**
[[{content of table 2}]]
```

The model outputs a per-column binary decision list indicating whether each column should be *directly concatenated* (0) or *semantically merged* (1). Direct concatenation applies when cells are cleanly split at the page boundary (e.g. a single cell’s content is broken across two rows), while semantic merging preserves both rows as distinct data. A typical semantic merging process is shown in Figure 11. This fine-grained,

Truncated Paragraph Merging

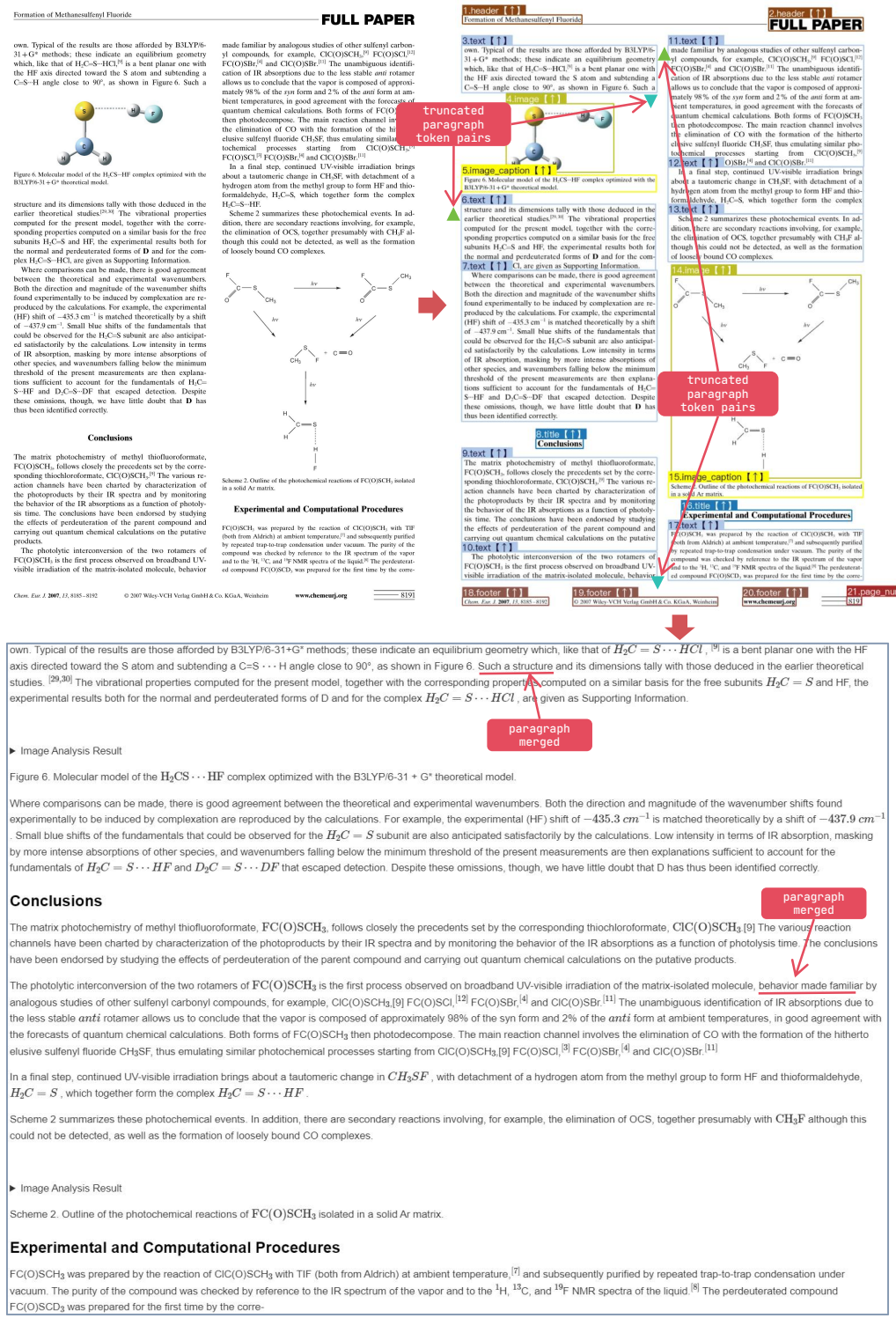


Figure 10: Illustration of truncated paragraph merging. In multi-column and complex layouts, Layout Detection splits continuous paragraphs into separate bounding boxes. The merge label predicted by the model reassembles them into coherent paragraphs during Markdown rendering.

Cross Page Table Recognition and Merging											
pre-page table					next-page table						
		实际空载开度, 观察机组转速上升平滑且不超过115%的额定转速 2.机组在额定转速运行一定时间后, 将机组各轴承温度稳定无其他异常现象后, 切换至自动位置, 此时接力器应无明显摆动, 在自动调节状态下, 其相对转速波动值不应超过0.20%	2、转速超过115%的额定转速扣5分; 3、未检查机组各轴承温度稳定无其他异常现象扣5分; 4、机组未稳定手动切换至自动位置扣5分;				跃变化4%的调速器调节品质要求: a) 频率变化衰减度不大于25%; b) 频率最大超调量不超过扰动量 Δf_0 的35%; c) 由扰动开始, 到调节稳定为止的调节时间 T_p 不超过25s; d) 在调节时间 T_p 内, 频率超过 $\pm 0.35\text{Hz}$ 的波动次数不超过两次。	$\pm 0.35\text{Hz}$ 的波动次数不超过两次扣3分。			
5	空载扰动试验	1、调速器调节参数进行核对; 2、设置一组调速器调节参数 (K_P, K_I, K_D), 观察空载稳定性与各项调节指标: a) 操作调速器电柜触摸屏, 置频给为51.00Hz, 机组稳定后置频给为49.00Hz, 记录频率最低值、稳定时间、调节次数; 稳定后将频给置51.00Hz, 记录频率最高值、调节次数、稳定时间。 b) 操作调速器电柜触摸屏置频给恢复50.00Hz。 c) 操作调速器电柜触摸屏, 置频给为52.00Hz, 机组稳定后置频给为48.00Hz, 记录频率最低值、稳定时间、调节次数; 稳定后将频给置52.00Hz, 记录频率最高值、调节次数、稳定时间。 d) 操作调速器电柜触摸屏置频给恢复50.00Hz。 3.重新一组调速器调节参数, 选择较好的调节参数 (K_P, K_I, K_D), 使调节品质达到最优, 频率阶	33	1. 未调速器调节参数进行核对扣3分; 2、设置一组调速器调节参数不全扣3分; 3、未进行2%的频率上阶跃扣3分; 4、未进行2%的频率下阶跃扣3分; 5、未进行4%的频率上阶跃扣3分; 6、未进行4%的频率下阶跃扣3分; 7、未置频给恢复50.00Hz扣3分; 8、选择调节参数, 调节品质未达到频率变化衰减度不大于25%扣3分; 9、选择调节参数, 调节品质未达到频率最大超调量不超过扰动量 Δf_0 的35%扣3分; 10、选择调节参数, 调节品质未达到调节时间 T_p 不超过25s扣3分; 11、选择调节参数, 调节品质未达到调节时间内频率超过			6	试验完毕	注消工作票、拆除试验接线、清理现场	9	1. 未注消工作票扣3分。 2. 未拆除试验接线扣3分。 3. 未清理现场扣3分。
							7	试验数据记录及分析	1.试验数据应包括试验时间、天气、试验主要仪器及精度、试验数据、试验人等 2.出具分析报告	10	1. 试验数据不全每项扣1分扣完5分为止。 2. 无出具分析报告扣5分。
8 安全风险扣分项											
8.1	严重违章否定项	参照水电专业红线条款	/	严重违章“一票否定”							
8.2	其他违章扣分	参照水电专业红线条款	/	一般违章按照10分/项次。							
考试开始时间						考试结束时间					
考试成绩						考评员		考评组长			

		实际空载开度, 观察机组转速上升平滑且不超过115%的额定转速2. 机组在额定转速运行一定时间后, 将机组各轴承温度稳定无其他异常现象后, 切换至自动位置, 此时接力器应无明显摆动, 在自动调节状态下, 其相对转速波动值不应超过0.20%	2、转速超过115%的额定转速扣5分;3、未检查机组各轴承温度稳定无其他异常现象扣5分;4、机组未稳定手动切换至自动位置扣5分;								
5	空载扰动试验	1. 调速器调节参数进行核对;2. 设置一组调速器调节参数((K_P, K_I, K_D)), 观察空载稳定性与各项调节指标:a) 操作调速器电柜触摸屏, 置频给为51.00Hz, 机组稳定后置频给为49.00Hz, 记录频率最低值、稳定时间、调节次数; 稳定后将频给置51.00Hz, 记录频率最高值、调节次数、稳定时间。b) 操作调速器电柜触摸屏置频给恢复50.00Hz。c) 操作调速器电柜触摸屏, 置频给为52.00Hz, 机组稳定后置频给为48.00Hz, 记录频率最低值、稳定时间、调节次数; 稳定后将频给置52.00Hz, 记录频率最高值、调节次数、稳定时间。d) 操作调速器电柜触摸屏置频给恢复50.00Hz。3. 重新一组调速器调节参数, 选择较好的调节参数((K_P, K_I, K_D)), 使调节品质达到最优, 频率阶跃变化4%的调速器调节品质要求: a) 频率变化衰减度不大于25%;b) 频率最大超调量不超过扰动量 Δf_0 的35%;c) 由扰动开始, 到调节稳定为止的调节时间 T_p 不超过25s; d) 在调节时间 T_p 内, 频率超过 $\pm 0.35\text{Hz}$ 的波动次数不超过两次。	33	1. 未调速器调节参数进行核对扣3分;2. 设置一组调速器调节参数不全扣3分;3. 未进行2%的频率上阶跃扣3分;4. 未进行2%的频率下阶跃扣3分;5. 未进行4%的频率上阶跃扣3分;6. 未进行4%的频率下阶跃扣3分;7. 未置频给恢复50.00Hz扣3分;8. 选择调节参数, 调节品质未达到频率变化衰减度不大于25%扣3分;9. 选择调节参数, 调节品质未达到频率最大超调量不超过扰动量 Δf_0 的35%扣3分;10. 选择调节参数, 调节品质未达到调节时间 T_p 不超过25s扣3分;11. 选择调节参数, 调节品质未达到调节时间内频率超过 $\pm 0.35\text{Hz}$ 的波动次数不超过两次扣3分。			6	试验完毕	注消工作票、拆除试验接线、清理现场	9	1. 未注消工作票扣3分。2. 未拆除试验接线扣3分。3. 未清理现场扣3分。
7	试验数据记录及分析	1. 试验数据应包括试验时间、天气、试验主要仪器及精度、试验数据、试验人等 2. 出具分析报告				10		1. 试验数据不全每项扣1分扣完5分为止。2. 无出具分析报告扣5分。			
8 安全风险扣分项											
8.1	严重违章否定项	参照水电专业红线条款	/	严重违章“一票否定”							
8.2	其他违章扣分	参照水电专业红线条款	/	一般违章按照10分/项次。							
考试开始时间						考试结束时间					
考试成绩						考评员		考评组长			

Figure 11: Cross-page table merging example. The model performs semantic understanding at the junction of two table fragments identified by rule-based heuristics, producing per-column merge decisions to reconstruct the complete table.

column-level strategy handles the common case where some columns require concatenation and others do not within the same table split.

In-table image detection. Tables in real-world documents frequently contain embedded images (e.g., product photos, diagrams, icons). MinerU2.5-Pro detects these through a three-step process:

1. **Detection.** Layout Detection identifies image regions that fall spatially within a table bounding box. Each detected in-table image is replaced with a special placeholder token in the table crop, effectively masking the image region.
2. **Recognition.** The masked table image is fed to Table Recognition, which generates the OTSL sequence with placeholder tokens marking the positions of masked images.
3. **Restoration.** In the final output, placeholder tokens are resolved back to references to the original image regions, producing HTML table cells that contain `<img>` tags with unique identifiers linking to the extracted image content blocks.

This approach allows the table structure and textual content to be recognized without interference from embedded images, while preserving the spatial correspondence between images and their containing cells in the final output. Representative examples are shown in [Figure 12](#).

C OmniDocBench v1.6 Detailed Results

This section reports per-model detailed results on the OmniDocBench v1.6 Base and Hard subsets. The evaluation protocol and matching corrections are described in [Section 5](#).

C.1 Detailed Results on Base Subset

[Table 7](#) reports detailed evaluation results for all models on the OmniDocBench v1.6 Base subset.

On the Base subset, top model scores are tightly clustered: the top 6 Overall scores fall within 94.49–96.19, a range of only 1.70 points. MinerU2.5-Pro ranks second at 96.12, only 0.07 points behind GLM-OCR, while leading in Text Edit Distance (0.033 *vs.* 0.039), Table TEDS (94.49 *vs.* 94.36), and Reading Order (0.109 *vs.* 0.122).

C.2 Detailed Results on Hard Subset

[Table 8](#) reports detailed evaluation results for all models on the OmniDocBench v1.6 Hard subset.

Rankings on the Hard subset differ markedly from Base, validating its effectiveness in differentiating model capabilities. Key observations: (1) MinerU2.5-Pro leads at 94.08, ahead of the second-place PaddleOCR-VL (92.48) by 1.60 points, achieving the best scores in Formula CDM (97.54), Table TEDS (89.91), and Reading Order (0.170). (2) GLM-OCR ranks first on Base (96.19) but drops to third on Hard (92.01), a decline of 4.18 points; HunyuanOCR drops from Base (92.45) to Hard (82.69), a decline of 9.76 points. In contrast, MinerU2.5-Pro declines by only 2.04 points, demonstrating the strongest robustness. (3) Gemini 3 Pro/Flash perform well on Hard (91.99), narrowing the gap with specialized models, benefiting from their strong capabilities on hard formulas (CDM 97.23/96.68).

D Qualitative Comparison with SOTA Methods

This section presents qualitative comparisons of parsing results between MinerU2.5-Pro and current SOTA methods on representative scenarios.

Table With Image									
	验证时间	注塑件批次	环境		辅助流程	油墨	钢板	熔接线状态	产品图片
条件二	2024年12月28日9:30-10:08	新产品 2024年12月24日	温度: 21.9℃ 湿度: 18%		热风枪	新油墨	新钢板 (35μm)	OK	
条件三	2024年12月28日9:30-10:08	新产品 2024年12月24日	温度: 21.9℃ 湿度: 18%		无	新油墨	新钢板 (35μm)	NG	
条件五	2024年12月28日13:30-14:00	老产品 2024年12月6日	温度: 23.6℃ 湿度: 16%		无	L: 新油墨 R: 上批次油墨	新钢板 (35μm)	L: OK R: NG	
条件六	2024年12月28日14:00-14:20	老产品 2024年12月6日	温度: 23.6℃ 湿度: 16%		无	R: 新油墨	新钢板 (35μm)	R: OK	

	验证时间	注塑件批次	环境		辅助流程	油墨	钢板	熔接线状态	产品图片
条件二	2024年12月28日9:30-10:08	新产品 2024年12月24日	温度: 21.9℃ 湿度: 18%		热风枪	新油墨	新钢板 (35μm)	OK	
条件三	2024年12月28日9:30-10:08	新产品 2024年12月24日	温度: 21.9℃ 湿度: 18%		无	新油墨	新钢板 (35μm)	NG	
条件五	2024年12月28日13:30-14:00	老产品 2024年12月6日	温度: 23.6℃ 湿度: 16%		无	L: 新油墨 R: 上批次油墨	新钢板 (35μm)	L: OK R: NG	
条件六	2024年12月28日14:00-14:20	老产品 2024年12月6日	温度: 23.6℃ 湿度: 16%		无	R: 新油墨	新钢板 (35μm)	R: OK	

UNSDGS	THE GROUP'S CONTRIBUTIONS
	Goal 3. Ensure healthy lives and promote well-being for all at all ages > The Group does not tolerate any form of harassment in the workplace > Provide employee health benefits, sports and recreational activities, compassionate leaves, maternal and paternal leaves > Reduced Occupational risks
	Goal 4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all > Invest in workforce education, training, and technical programmes
	Goal 5. Achieve gender equality and empower all women and girls > Upholds the practice of equal opportunity to its employees, clients, suppliers and other stakeholders
	Goal 8. Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all > Conduct skills assessment and communicate reasonable expectations > Foster productive local employment and workforce development > Encourage local procurement and supplier development > Established Human rights and labour Policy
	Goal 9. Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation > Upgrade infrastructure and technology as well as enhance technological capabilities > Started the digitalisation programme
	Goal 12. Ensure sustainable consumption and production patterns > Practice environmentally sound waste management and recycling activity > Monitor & reduce wastage
	Goal 13. Take urgent action to combat climate change > Disclosed GHG emission according to GHG Protocol, and set GHG emission target > In this report we included an initial statement to Task Force on Climate-related financial disclosure.
	Goal 14. Conserve and sustainably use oceans, seas and marine resources for sustainable development > Incorporate environmental assessments into management plans > Implement environmental accident prevention, preparedness and response
	Goal 16. Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels > The Group Policies address Conflicts of Interest and Related Party Transactions, as well as Bribery, Corruption, Business conduct and ethics. > Conduct stakeholder engagement between the public and private sectors > Enhanced risk management

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Figure 12: In-table image detection and recognition. Embedded images within table cells are masked with placeholders during Table Recognition and restored as references in the final HTML output.

Table 7: Detailed results on OmniDocBench v1.6 Base subset.

Model	Overall↑	Text Edit↓	Formula CDM↑	Table TEDS↑	TEDS-S↑	Read Order↓
GLM-OCR	96.19	0.039	98.10	94.36	96.59	0.122
MinerU2.5-Pro	96.12	0.033	97.16	94.49	96.63	0.109
PaddleOCR-VL-1.5	95.72	0.032	97.18	93.17	95.31	0.118
Youtu-Parsing	94.87	0.038	94.32	94.07	96.56	0.101
Ovis2.6-30B-A3B	94.56	0.031	95.11	91.64	94.33	0.125
PaddleOCR-VL	94.49	0.035	95.43	91.51	94.39	0.123
Logics-Parsing-v2	94.16	0.036	95.06	91.04	93.92	0.127
FireRed-OCR	94.14	0.030	95.63	89.76	92.41	0.120
MinerU2.5	93.23	0.042	94.81	89.06	92.24	0.120
OpenDoc-0.1B	93.04	0.039	94.05	88.93	91.99	0.126
Gemini 3 Pro	92.96	0.060	95.11	89.80	93.36	0.157
Gemini 3 Flash	92.58	0.062	94.19	89.74	93.82	0.163
HunyuanOCR	92.45	0.082	91.09	94.44	95.76	0.156
DeepSeek-OCR-2.0	91.50	0.046	93.02	86.13	89.75	0.134
dots.ocr	90.91	0.041	88.85	88.02	90.99	0.126
Dolphin-2.0	90.42	0.064	89.98	87.67	90.31	0.137
Qwen3-VL-235B	90.08	0.062	91.86	84.61	87.89	0.157
OCRverse	89.36	0.054	88.77	84.67	88.19	0.152
MonkeyOCR-pro-3B	89.15	0.067	86.87	87.22	90.49	0.181
Dolphin-1.5	87.24	0.091	86.32	84.47	87.55	0.157
GPT-5.2	86.83	0.120	88.62	83.82	88.31	0.188
Mistral-OCR	86.36	0.095	89.28	79.34	83.32	0.161
POINTS-Reader	86.20	0.095	89.69	78.37	81.43	0.184
Nanonets-OCR-s	86.10	0.099	86.43	81.75	85.41	0.192
olmOCR	85.89	0.135	87.00	84.17	87.64	0.205
InternVL 3.5	83.76	0.137	89.39	75.58	80.53	0.214

D.1 Table Recognition

MinerU2.5-Pro demonstrates superior accuracy on complex tables, particularly rotated tables and tables with long merged cells. As shown in [Figure 13](#) and [Figure 14](#), MinerU2.5-Pro correctly recovers the table structure and content, while competing models exhibit noticeable structural errors such as misaligned rows and lost cell boundaries.

D.2 Formula Recognition

The decoupled row-by-row formula analysis of MinerU2.5-Pro yields high accuracy on multi-line formulas, substantially outperforming end-to-end approaches that must generate entire equation groups in a single pass. MinerU2.5-Pro also achieves more accurate recognition on complex matrices. Representative comparisons are shown in [Figure 15](#) and [Figure 16](#).

D.3 Image-Aware Parsing

MinerU2.5-Pro’s image-aware parsing extracts structured content from chart and diagram regions that other models typically leave as opaque image placeholders. [Figure 17](#) and [Figure 18](#) compare parsing results across different chart types.

Table 8: Detailed results on OmniDocBench v1.6 Hard subset.

Model	Overall↑	Text Edit↓	Formula CDM↑	Table TEDS↑	TEDS-S↑	Read Order↓
MinerU2.5-Pro	94.08	0.052	97.54	89.91	93.61	0.170
PaddleOCR-VL	92.48	0.066	96.24	87.84	91.60	0.189
GLM-OCR	92.01	0.066	94.81	87.81	91.44	0.186
PaddleOCR-VL-1.5	92.01	0.065	95.74	86.75	91.30	0.181
Gemini 3 Flash	91.99	0.085	96.68	87.83	92.51	0.214
Gemini 3 Pro	91.99	0.083	97.23	87.03	91.68	0.198
MinerU2.5	91.65	0.062	97.12	84.00	88.97	0.178
Ovis2.6-30B-A3B	90.39	0.056	94.56	82.21	86.07	0.184
Logics-Parsing-v2	89.95	0.062	96.28	79.81	85.61	0.184
FireRed-OCR	89.89	0.073	94.57	82.40	86.64	0.183
Youtu-Parsing	89.81	0.076	91.75	85.30	89.90	0.185
dots.ocr	88.67	0.081	89.65	84.42	89.25	0.196
Qwen3-VL-235B	88.45	0.065	93.85	78.01	83.00	0.210
DeepSeek-OCR-2.0	86.23	0.067	88.81	76.53	81.19	0.191
GPT-5.2	86.07	0.087	86.79	80.10	86.70	0.213
Dolphin-2.0	85.29	0.094	91.61	73.68	78.04	0.210
MonkeyOCR-pro-3B	85.07	0.109	91.18	74.92	82.48	0.228
OCRverse	84.79	0.106	89.86	75.11	79.99	0.215
olmOCR	84.34	0.157	89.58	79.17	85.65	0.269
InternVL 3.5	83.44	0.098	89.77	70.30	77.35	0.222
Dolphin-1.5	83.38	0.106	89.30	71.43	75.86	0.215
Mistral-OCR	82.85	0.104	90.62	68.36	73.08	0.219
OpenDoc-0.1B	82.69	0.100	90.73	67.32	72.55	0.206
HunyuanOCR	82.69	0.120	80.32	79.76	84.92	0.243
Nanonets-OCR-s	76.90	0.154	70.99	75.05	81.56	0.309
POINTS-Reader	74.86	0.103	75.23	59.60	64.19	0.263

MinerU2.5-Pro: Pushing the Limits of Data-Centric Document Parsing at Scale

[illegible]

Paddle-VL1.5

中国医美市场趋势洞察报告 | 第七部分：医美行业主要上市公司财务情况

单位：人民币百万元

产业链位置	公司名称	代码	医美产品	上市时间	营业收入	净利润	毛利率	净利率	市值	市盈率	2018年12月28日	2019年12月31日	2020年6月30日

资料来源：Capital IQ、Wind数据库

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MinerU2.5-Pro

中国医美市场趋势洞察报告 | 第七部分：医美行业主要上市公司财务情况

单位：人民币百万元

产业链位置	公司名称	代码	医美产品	上市时间	营业收入			净利润			毛利率			净利率			市值			市盈率			
					2018	2019	2020年1-9月	2018	2019	2020年1-9月	2018	2019	2020年1-9月	2018	2019	2020年1-9月	2018年12月31日	2019年12月31日	2020年6月30日	2018	2019	2020年1-9月	
上游	原材料及产品提供商	华熙生物	688363.CH	玻尿酸填充剂+“童颜针”注射用玻尿酸	06/11/2019	1,263	1,886	1,596	424	586	437	80%	80%	81%	34%	31%	27%	40,832	70,992		68	73	
	原材料及产品提供商	昊海生物	602616.N/605366.CH	玻尿酸填充剂+“海得”注射用玻尿酸	30/04/2015	1,546	1,595	887	415	371	102	79%	77%	77%	27%	23%	12%	5,880	14,821	13,779	14	38	44
	原材料及产品提供商	爱美客	300898.CH	玻尿酸填充剂	28/09/2020	321	558	464	123	306	206	89%	93%	92%	38%	55%	62%						69
	原材料及产品提供商	科韩生物	1786.TT	玻尿酸填充剂+“童颜针”注射用玻尿酸	29/12/2010	76	102	72	22	31	19	66%	70%	68%	28%	31%	26%	904	1,020	924	43	33	111
	原材料提供商	阜丰集团	52HK546	玻尿酸填充剂	08/02/2007	13,909	16,346	16,636	1,845	1,137	904	19%	20%	19%	13%	7%	5%	6,601	7,881	5,696	4	7	3
	产品提供商	双鹭药业	002038.CH	子公司生产“野韵”注射用玻尿酸	27/01/2005	2,167	2,030	842	571	487	302	79%	79%	89%	26%	24%	36%	25,480	13,926	13,156	45	28	20
中游	产品提供商	聚嘉生物	300238.CH	玻尿酸的原料	06/07/2011	408	438	319	46	(465)	36	73%	68%	73%	10%	-106%	11%	2,528	5,064	4,700	55	(11)	82
	医美机构	朗姿股份	002611.CH	韩翰韩熙熙医疗美容集团	30/08/2011	2,662	3,007	1,997	210	59	48	58%	58%	54%	8%	2%	2%	3,881	4,296	4,500	18	73	41
	医美机构	康美医疗集团	Nasdaq:GMAH	玻尿酸填充剂	25/10/2019	761	869	777	(255)	136	(216)	66%	68%	61%	-34%	16%	-28%		1,055	1,030	-	8	(2)
	医美机构	华韩整形	430333.OC	玻尿酸医美机构	06/11/2013	693	804	4	8	83		51%	51%		1%	10%		445	1,002	1,440	125	12	
	医美机构	利美泰	832533.OC	玻尿酸医美机构	04/06/2015	315	153	(15)	(69)			51%	33%	-15%	-45%			402	321	174	(27)	(5)	
	医美机构	埃维诺美(已退市)	832605.OC	玻尿酸医美机构	26/06/2016	70		(13)				50%			-19%								
下游	医美电商平台	新氧	SYUS	线上医美服务	02/05/2019	617	1,152	870	55	177	(33)	85%	83%	83%	9%	15%	-4%	6,071	7,199		34	(20)	

资料来源：Capital IQ、Wind数据库

Figure 13: Qualitative comparison on rotated table recognition. MinerU2.5-Pro correctly recovers the rotated structure, while competing models produce misaligned rows or missing cells.

MinerU2.5-Pro: Pushing the Limits of Data-Centric Document Parsing at Scale

Image

Spec. No. JEN-PN2A3A 0033AA-01

Reference Only

P.2/12

Customer Part Number	MURATA Part Number	Impedance (Ω) (at 100MHz, Under Load Testing Condition)	Rated Current (mA) [85°C, 125°C]	DC Resistance		Remark
				Initial Values	Values After Testing	
	BLM18AG1T1-9NB10	120±20%	120	800	0.18	0.28
	BLM18AG1J1-9NB10	150±20%	150	700	0.25	0.35
	BLM18AG1S1-9NB10	200±20%	200	700	0.25	0.35
	BLM18AG2T1-9NB10	330±20%	330	600	0.30	0.40
	BLM18AG2J1-9NB10	470±20%	470	550	0.35	0.45
	BLM18AG2S1-9NB10	600±20%	600	500	0.38	0.48
	BLM18AG4T1-9NB10	1000±20%	1000	450	0.50	0.60
	BLM18H800S0-9NB10	5±25%	5	800	0.05	0.10
	BLM18H800J0-9NB10	5±25%	5	900	0.2	0.3
	BLM18H800S1-9NB10	10±25%	10	700	0.10	0.20
	BLM18H800J1-9NB10	10±25%	10	500	0.25	0.35
	BLM18H800S2-9NB10	22±25%	22	700	0.30	0.30
	BLM18H800J2-9NB10	22±25%	22	900	0.38	0.48
	BLM18H84T0-9NB10	47±25%	47	600	0.25	0.35
	BLM18H84J0-9NB10	47±25%	47	500	0.3	0.4
	BLM18H84T0S0-9NB10	47±25%	47	300	0.05	0.05
	BLM18H84T0J0-9NB10	60±25%	60	600	0.25	0.35
	BLM18H84T0S1-9NB10	75±25%	75	300	0.70	0.80
	BLM18H84T0J1-9NB10	75±25%	75	600	0.30	0.40
	BLM18H870S1-9NB10	120±25%	120	550	0.30	0.40
	BLM18H80T1-9NB10	120±25%	120	300	0.4	0.5
	BLM18H84T1-9NB10	150±25%	150	700	n.a	0.1
	BLM18H81T1-9NB10	140±25%	140	500	0.35	0.45
	BLM18H81J1-9NB10	150±25%	150	450	0.37	0.47
	BLM18H80S1-9NB10	150±25%	150	300	0.4	0.5
	BLM18H80J1-9NB10	220±25%	220	450	0.45	0.55
	BLM18H80S2-9NB10	220±25%	220	250	0.45	0.55

MURATA MFG. CO., LTD.

MinerU2.5-Pro

Reference ID#:

Spec. No. SINC-CA-1000A-01

P.02/02

Customer Part Number	MINER Part Number	Impedance 25 °C (100kΩ, 100k Standard Testing Condition)	Rated Load Power		DC Resistance @ 25 °C		Remark
			Rated	100%	MINER	MINER (After Solder)	
	BSF0000000000000000	100kΩ/2W	100	100	0.15	0.15	Not general use
	BSF0000000000000001	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000002	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000003	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000004	100kΩ/2W	100	100	0.15	0.15	For high speed signal line
	BSF0000000000000005	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000006	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000007	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000008	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000009	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000010	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000011	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000012	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000013	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000014	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000015	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000016	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000017	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000018	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000019	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000020	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000021	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000022	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000023	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000024	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000025	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000026	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000027	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000028	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000029	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000030	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000031	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000032	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000033	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000034	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000035	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000036	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000037	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000038	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000039	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000040	100kΩ/2W	100	100	0.15	0.15	
	BSF0000000000000041	1					

MINER-MINIC-001

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Figure 14: Qualitative comparison on tables with long merged cells. MinerU2.5-Pro preserves the span structure, whereas other models incorrectly split or duplicate merged cells.

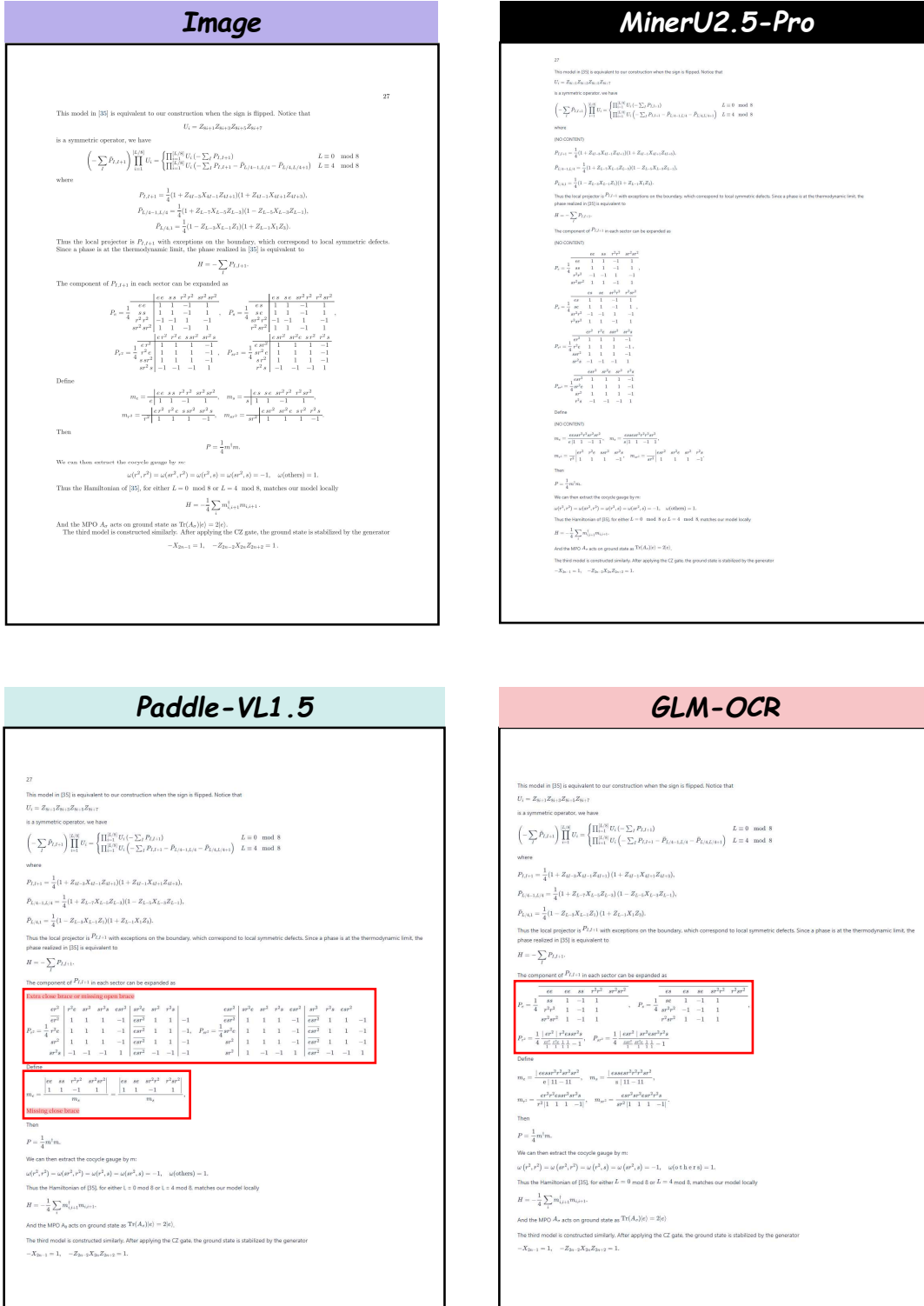


Figure 15: Qualitative comparison on complex matrix recognition. MinerU2.5-Pro accurately captures the matrix structure and alignment, while other models exhibit symbol errors or structural collapse.

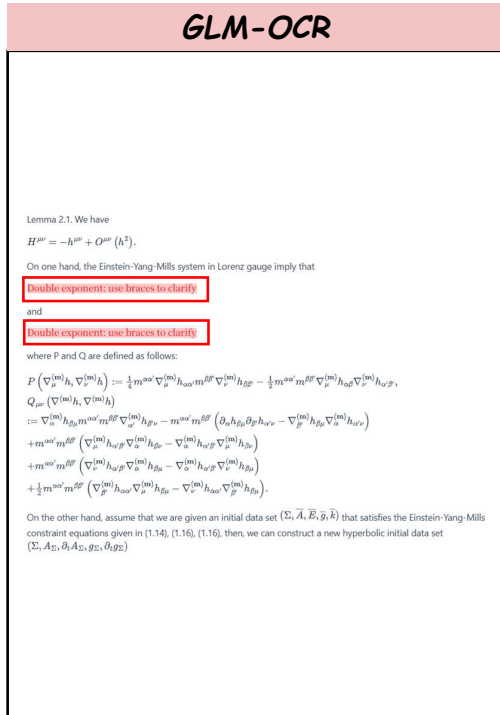
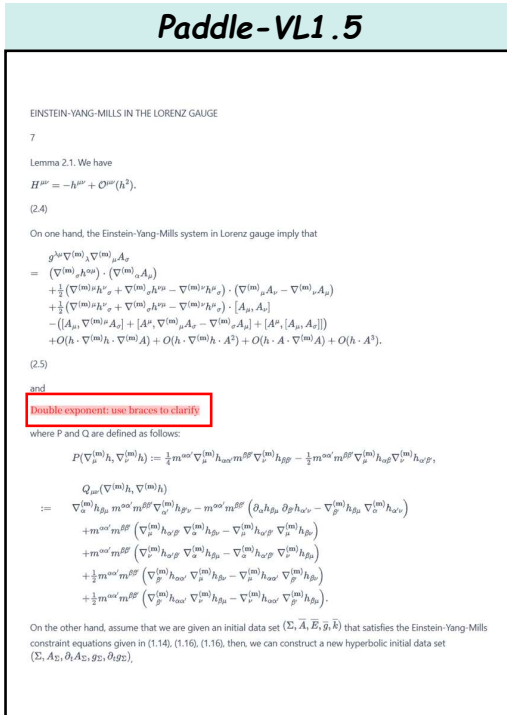
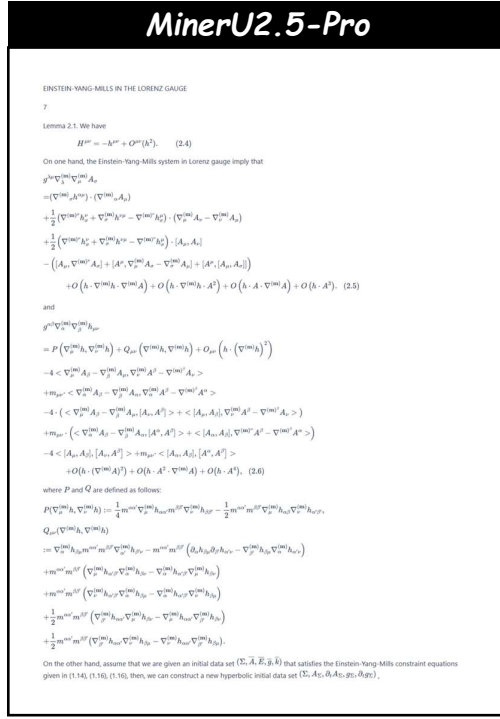
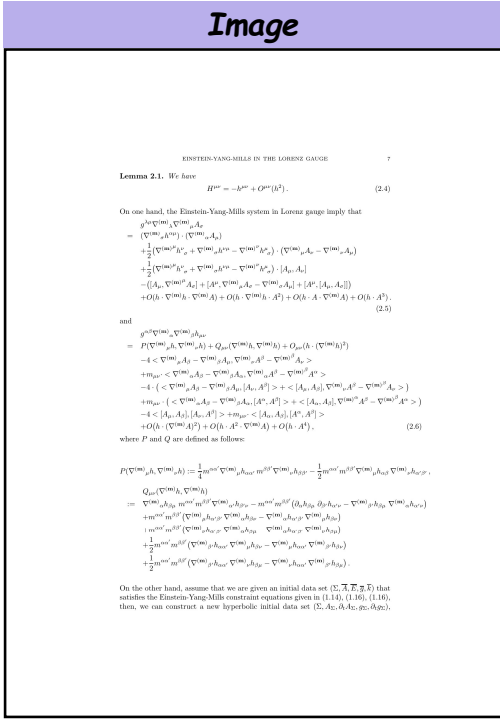


Figure 16: Qualitative comparison on multi-line formula recognition. The row-by-row analysis of MinerU2.5-Pro faithfully reproduces each line, whereas competing models merge or misalign lines.

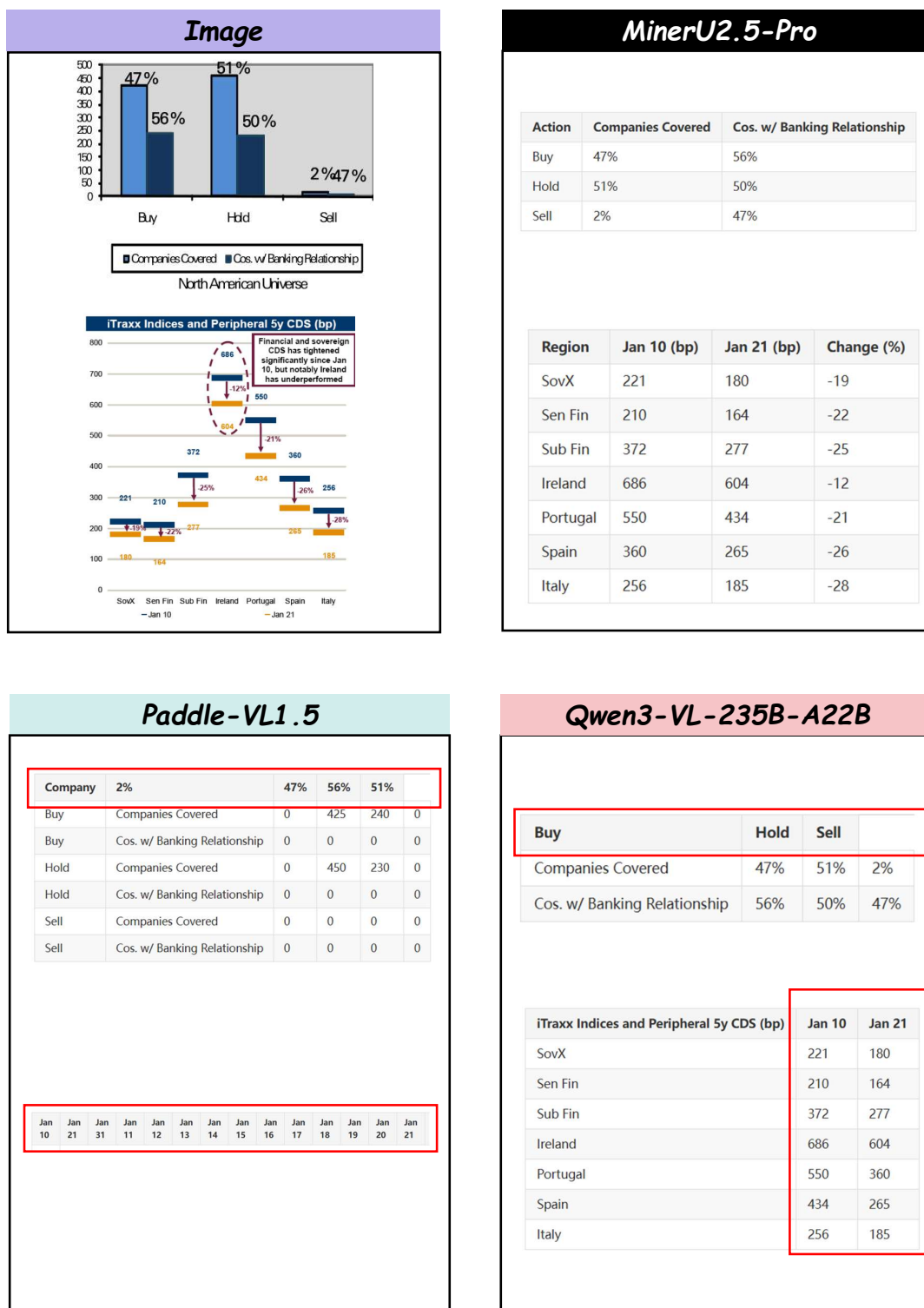


Figure 17: Qualitative comparison on image-aware chart parsing (Part 1). MinerU2.5-Pro extracts structured content from diverse chart types, while other models either ignore or misinterpret chart content.

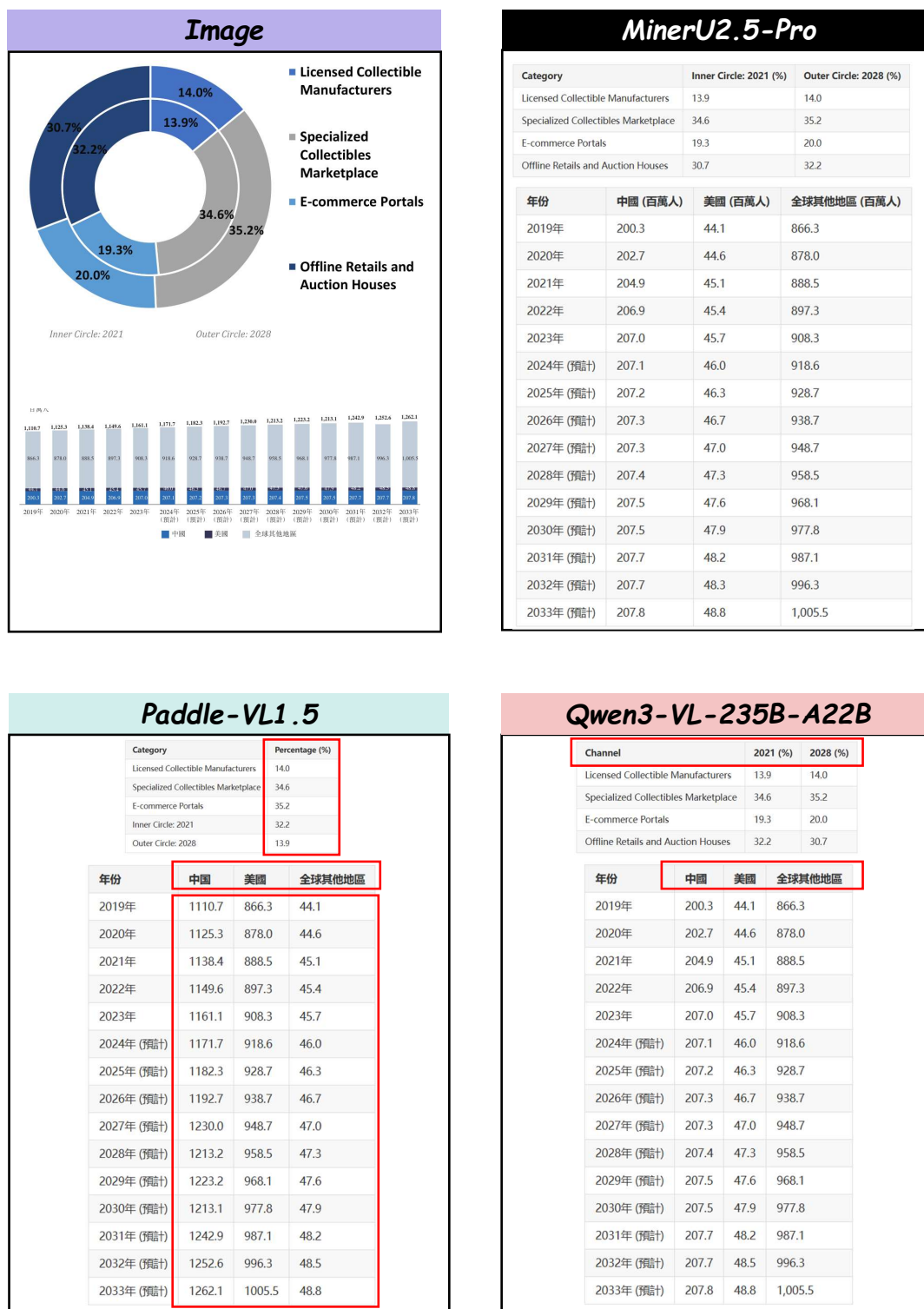


Figure 18: Qualitative comparison on image-aware chart parsing (Part 2). Additional chart types demonstrating the generalization of MinerU2.5-Pro's image analysis pipeline.